Energy & Buildings 360 (2026) 117425

Contents lists available at ScienceDirect

Energy & Buildings

journal homepage: www.elsevier.com/locate/enb

# State disaggregation for Non-Intrusive Load Monitoring (NILM) with calibrated bagged Seq2Point 1D convolutional neural networks

Md Mahadi Hasan Imran [a], Muhamad Zalani Daud [a,b,*], Muhammad Mukhtar Qureshi [c], Shahrizan Jamaludin [a], Atta-ur-Rahman [d], Md Meherullah [a], Abdullah Alqahtani [e]

[a] Faculty of Ocean Engineering Technology, Universiti Malaysia Terengganu, 21030 Kuala Nerus, Terengganu, Malaysia
[b] Centre for Offshore Renewable Energy, Universiti Malaysia Terengganu, 21030 Kuala Nerus, Terengganu, Malaysia
[c] Department of Smart Computing and Cyber Resilience (DSCCR), School of Computing and Artificial Intelligence, Faculty of Engineering and Technology, Sunway University, No. 5, Jalan Universiti, Bandar Sunway, 47500 Selangor, Darul Ehsan, Malaysia
[d] Department of Computer Science, College of Computer Science and Information Technology, Imam Abdulrahman Bin Faisal University, P.O. Box 1982, Dammam 31441, Saudi Arabia
[e] Department of Computer Information Systems, College of Computer Science and Information Technology, Imam Abdulrahman Bin Faisal University, P.O. Box 1982, Dammam 31441, Saudi Arabia

ARTICLE INFO

*Keywords:*
Non-intrusive load monitoring (NILM)
Seq2Point CNN
Ensemble learning
Energy efficiency
Machine learning
Energy disaggregation
Electrical power system

ABSTRACT

Non-intrusive load monitoring (NILM) aims to infer appliance operating states from aggregate power data, yet reliable state disaggregation remains challenging because appliance labels are often generated using heuristic thresholds, ON/OFF classes are highly imbalanced, prediction probabilities are poorly calibrated, and output sequences can become fragmented and unstable. To address these limitations, this research proposes a calibrated bagged Seq2Point one-dimensional convolutional neural network, termed CB-S2P NILM. The framework integrates validation-driven per-device threshold optimization, class-balanced binary cross-entropy, temperature scaling, temporal smoothing, and dwell-time constraints, together with bagged ensemble prediction to improve robustness and predictive stability. Evaluated on the REDD dataset across six appliances, the proposed method demonstrated strong and consistent performance. In the validation phase, the Bagged CNN achieved appliance-level accuracies of 0.98 to 0.99 and F1-scores of 0.98 to 0.99, outperforming the baseline CNN across all evaluated appliances. Sensitivity analysis further showed that an ensemble size of three yielded the best overall performance, with an average accuracy of 0.99 and an average F1-score of 0.99. Overall, the proposed CB-S2P NILM provides a robust, interpretable, and reproducible framework for appliance-state disaggregation within the evaluated REDD cross-house setting.

## 1. Introduction

Non-intrusive load monitoring (NILM) separates the contribution of individual appliances from a single aggregate electricity signal, enabling end-use visibility without installing dedicated sensors on every device [1–3]. In practical settings such as demand response, fault diagnosis, and home energy automation, the main requirement is often dependable identification of appliance activation patterns, since knowing when a device turns on and off is more useful for decision-making than reconstructing a detailed power waveform [3,4]. State disaggregation therefore addresses the central question of when an appliance was operating and supports interpretable analytics such as duty-cycle estimation, duration analysis, and transition timing.

However, generating stable and trustworthy appliance state sequences from aggregate measurements remains a persistent challenge [5,6]. In many NILM studies, training labels are derived from ad hoc power thresholds that vary across devices and datasets, which introduces noisy supervision into model development [6]. At the same time, NILM data are naturally imbalanced because appliances remain OFF for much longer periods than they remain ON, while overlapping appliance operation further complicates signal separation [7]. Under these conditions, deep learning models often produce miscalibrated confidence scores, making fixed decision thresholds unreliable, and minute-level predictions frequently exhibit temporal chatter, fragmenting genuine activation periods into several spurious short runs [6,8–11]. These difficulties indicate that a practical NILM framework

* Corresponding author.
*E-mail address:* zalani@umt.edu.my (M.Z. Daud).

https://doi.org/10.1016/j.enbuild.2026.117425
Received 21 October 2025; Received in revised form 24 March 2026; Accepted 31 March 2026
Available online 1 April 2026
0378-7788/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

should not only be accurate, but also label-aware, statistically calibrated, temporally stable under low-frequency operating conditions.

Recent NILM research has demonstrated that strong representation learning can achieve excellent results when high-frequency voltage-current (V-I) measurements are available. Event-centric approaches inspired by object detection frameworks, such as DeepDFML-NILM [12], jointly localize events and classify appliances with high accuracy on curated datasets. Similarly, image-like or trajectory-based representations of short V-I windows have been widely adopted to strengthen appliance discrimination. Ma et al. [13] combined cumulative sum detection with V-I image encoding and reported near-ceiling classification performance, while Chen et al. [14] introduced a two-stream CNN using temporal and spectral Gramian fields to improve feature diversity. Related studies further explored twin CNNs over harmonic maps [15], learnable image-like encodings before 2D-CNN classification [16], quantized FPGA-oriented CNN implementations [17], RGB heatmap representations [18], and scattering-transform features coupled with classical machine learning classifiers [19]. Monteiro et al. [20] likewise showed that CNNs applied directly to high-frequency current signals can provide strong performance with reduced preprocessing demands. Collectively, these studies confirm the effectiveness of rich transient-level representation learning. However, they are typically built around waveform-rich sensing conditions, single-event assumptions, or highly curated experimental settings, and they do not directly resolve the problem of stable multi-appliance state disaggregation under minute-level residential smart-meter data [12,17,18,20–22].

For low-frequency NILM, where only aggregate active-power streams are available, sequence learning has become the dominant paradigm. Compact Seq2Point one-dimensional CNNs provide strong baselines by mapping a temporal window to a central target label, while recurrent, hybrid, and attention-based extensions have attempted to improve temporal dependency modeling. Belguermi et al. [23] showed that a 1D CNN-LSTM hybrid can achieve high F1-scores on AMPds2, although performance remained weaker for low-power and composite loads. Grover et al. [24] reported gains with attention-augmented convolutional recurrent designs, and transformer-style models proposed by Lin et al. [25] and Cheng and Zhong [26] further improved some appliance-level results. Other low-frequency studies have explored LSTM-attention architectures [27], multi-branch CNN-GRU attention models [28], and dual-task frameworks that jointly perform energy regression and state classification [29]. In parallel, some researchers converted minute-level series into visual forms to exploit 2D-CNNs, including temporal bar graphs [30] and vizgram matrices [31], while others enhanced temporal consistency through CNN-LSTM integration [32], de-noising autoencoder structures with feature modulation [33], convolutional autoencoders for regression-based disaggregation [34], composite loss design [35], or semi-supervised temporal convolutional learning [36]. Despite these methodological advances, the same practical limitations appear repeatedly across the literature: sensitivity to label quality, difficulty with low-power or polymorphic appliances, increased complexity in attention-based and hybrid architectures, and limited generalization across households [23,28,29,31,32,35].

Moreover, a broader comparison of recent NILM research reinforces these unresolved issues. Yan et al. [37] showed that well-designed statistical features combined with random forests can remain highly competitive in cross-household settings, suggesting that deeper architectures alone do not guarantee robust generalization. Wang et al. [38] proposed a pattern-based CNN for detecting rising, dropping, and normal states, but the framework did not support full multi-appliance disaggregation. Zhao et al. [39] investigated unsupervised state-transition modeling and achieved promising F-measures, though the method depended strongly on accurate segmentation and was less reliable under overlapping appliance activity. Sykiotis et al. [40] demonstrated that transformer pre-training can improve low-frequency NILM, but sparse activations remained challenging. Shahab et al. [41] combined Seq2Point learning with spectrogram-based CNN transfer learning and obtained solid results on REDD and REFIT, although synchronization and overlap limitations persisted. Domain-specific studies have also broadened NILM applications, including shipboard systems [42] and low-frequency event classification using active and reactive power [43]. Nevertheless, even advanced architectures such as Tra-CNN [26] and inception-style multiscale CNNs [44] continue to reveal trade-offs among model complexity, receptive-field design, and computational burden. Moreover, Table 1 provides a consolidated summary of recent NILM research, categorizing existing methods according to their core architectures, datasets, reported performance, and key limitations.

These collective findings make the current research gap more explicit. First, appliance state labels are still commonly generated using heuristic thresholds that are not systematically validated per device, thereby introducing label noise into both training and evaluation [23,30,31,35]. Second, although deep neural models often output highly confident probabilities, explicit probability calibration remains underused in NILM, which leaves decision thresholds brittle when appliance distributions shift [26,27,29,45]. Third, minute-level predictions frequently oscillate between ON and OFF states because temporal smoothing and dwell-time logic are either absent or weakly justified, leading to fragmented activations and inflated false transition rates [23,30,31]. Finally, NILM datasets are typically limited in size and highly heterogeneous, which makes predictions sensitive to random initialization and increases model variance, while ensemble bagging remains rarely explored for low-frequency state classification [24,46].

To address these limitations, this study proposes a Calibrated Bagged Seq2Point NILM framework (CB-S2P NILM) for state disaggregation under low-frequency smart-meter conditions without relying on heavy recurrent, attention-based, or time–frequency fusion architectures. The proposed framework combines validation-driven appliance-specific label calibration, imbalance-aware Seq2Point CNN learning, probability calibration through temperature scaling, temporally consistent post-processing, and bagged ensemble prediction within a unified design. In this way, the method is intended to improve the reliability, interpretability, and robustness of low-frequency NILM within the evaluated REDD cross-house setting.

The main contributions of this study are threefold. First, it introduces a validation-driven label construction strategy that replaces fixed heuristic thresholding with appliance-specific threshold optimization. Second, it develops a compact Seq2Point CNN framework for state disaggregation that explicitly incorporates imbalance-aware learning, calibrated decision-making, and temporal stabilization. Third, it demonstrates the use of bagged ensemble prediction as an effective variance-reduction strategy for improving robustness in low-frequency NILM classification.

The remainder of this paper is organized as follows. Section 2 presents the proposed methodology, including data preprocessing, label calibration, model architecture, probability scaling, temporal smoothing, and ensemble learning. Section 3 reports the experimental results on the REDD dataset, together with appliance-wise analysis and benchmarking against existing NILM approaches. Finally, Section 4 concludes the study, summarizes the main findings, and outlines directions for future research.

## 2. Methodology

### 2.1. Model development

The CB-S2P framework was designed to address several practical challenges of low-frequency NILM, including noisy appliance-state labeling, severe ON/OFF imbalance, unstable probability-to-state conversion, temporally fragmented predictions, and variance across model runs. Accordingly, the methodology combines feature-enhanced Seq2Point learning, imbalance-aware optimization, probability calibration, temporal post-processing, and bagged prediction so that each component addresses a distinct source of error in minute-level state

**Table 1**
Summary of recent NILM methods categorized by architecture type, dataset used, reported performance, and key limitations.

| Methods | Dataset(s) | Performance (as reported) | Key limitation(s) |
|---|---|---|---|
| High-frequency V–I/image CNNs: Two-Stream [13,14]; Twin CNN [15]; Learnable signatures [66]; Tiny V–I [67]; HF transients [68] | PLAID/ PLAID3, WHITED, COOLL; custom V–I sets | Acc often >95%; F1 ~0.92–0.99; class bests up to ~99.7 | Requires HF voltage/ current; assumes isolated/start-up events; weak cross-house generalization; hyperparameter sensitivity; event-only focus (no full disaggregation) |
| Low-frequency Seq2Point/CNN (state or regression): Edge S2P [69]; CNN-LSTM-RF [70]; DCML [71] | UK-DALE, REDD, REFIT, AMPds | MAE much lower than AFHMM; Acc ~93–98%; F1 up to ~0.99; energy error ~10–12% (edge) | Needs per-appliance/ window tuning; overlapping/low-power loads harder; some works lack calibration and temporal smoothing; appliance-specific training common |
| RNN/GRU/LSTM & CNN hybrids: LSTM + Bayes [58]; LSTM-CNN [23]; MCSTFF-A [28]; TCN-LSTM [72] | REDD, AMPds2, UK-DALE, REFIT, Pecan Street | Acc ~93–99%; F1 ~0.69–0.99; MAE/ RMSE improved vs baselines | High compute/ parameters; class-imbalance sensitivity; weaker on short/bursty or composite loads; per-appliance tuning and generalization limits |
| Transformer/ Attention hybrids: TransUNet-NILM [49,71]; Tra-CNN [26]; | REDD, UK-DALE, REFIT | F1 commonly ~0.98–0.99 | High computational cost/attention overhead; needs large data; instability on long sequences; weaker on small/ low-power loads; little calibration use |
| Domain adaptation/ Transfer learning: Seq2[3]Point + TL [41,73]; Diffusion TL [50]; TCN + KD + GRL [74]. | REDD, REFIT, UK-DALE | Acc ~86–95%; F1 ~0.42–0.94; sometimes + 5–20% with TL | Domain shift remains; per-appliance retraining often needed; complex training; calibration/temporal stability rarely addressed |
| Classical features + ensembles: VMD ensemble [75]; WRG + CNN [76–79]; RP/ GADF + CNN [80]; DCNN trajectory [81]. | PLAID, WHITED, COOLL, REDD | Acc up to 99.6–99.9%; F1 up to ~99.7 on sub-metered/HF | Dataset-specific features; assumes single switching; limited aggregate NILM validation; brittle without careful tuning |
| Event detection (rise/drop, event-only): BLUED LSTM/CNN [82]; VIIF-TSDM [83]. | AMPds, BLUED, PLAID/ QLTD | Recall ≈99% (patterns); F1 ≈96–99 (events) | Not full per-appliance disaggregation; minute-level misses fast transients; predetectors add false positives; single-house validations |
| Unsupervised/Self-supervised/Auto-labeling: SNN + STDP, CNN + NLSTM [84–86]; auto-label NILM, Hierarchical Decision-Making, CNN-BiLSTM, 3D-CNN [87,88] | PLAID; UK-DALE; REDD + private CN | Acc/F1 ~83–96 depending on setup; labeling improved | Lower than supervised CNNs; depends on knowledge bases or heavy pretraining; manual tie-breaks; hardware/ implementation complexity |
| Edge/compression/ pruning: Edge optimization, ELECTRIcity [40,89]; OPT-NILM pruning [90,91]; FPGA CNN [17]; Tiny V–I [67]. | UK-DALE, REDD, BLUED, COOLL | 75% storage reduction; similar MAE at 20–95%; sparsity; ~21 FPS on FPGA; ~98% accuracy (tiny V–I) | Risk of accuracy loss if over-compressed; per-appliance retraining; hardware limits; serial inference bottlenecks |
| Graph/ hierarchical/ statistical: GNN clustering [92]; TreeCNN [93]. | UK-DALE, REDD; Dataport (hourly) | Acc ~93; F1 ~0.88; MAE best among hourly baselines | High complexity; interpretability/ training stability issues; hourly resolution limits appliance dynamics |

disaggregation.

### 2.1.1. Data cleaning, feature construction, and windowing

In low-frequency NILM, the aggregate mains signal is often affected by missing samples, outliers, and overlapping appliance activity, which can obscure the signatures needed for state discrimination. Therefore, data cleaning, feature construction, and Seq2Point windowing are introduced to stabilize the aggregate representation while preserving the local temporal context required for appliance-state prediction.

A strictly monotone time index {t = 1,…,T} is first established and duplicate rows are removed to ensure temporal consistency. Short missing segments of up to five minutes are imputed by forward filling to preserve continuity, whereas longer gaps are explicitly masked and excluded from all downstream windowing so that no artificial information is introduced. Moreover, basic physical sanity checks are then applied to the aggregate power signal. Negative power readings are clipped to zero, and rare, implausible spikes are winsorized at the 99.9th percentile. This treatment prevents such outliers from dominating subsequent normalization and stabilizes training.

Feature construction is performed solely from the aggregate mains according to the definitions in Eq. (1) through Eq. (4). Four input channels are produced from these definitions, using fixed window sizes w ∈ {5,7} minutes that are held constant across all data splits. For robustness, the location–scale operators $\mu^{(w)}$ and $\sigma^{(w)}$ could be replaced by a median and median absolute deviation; however, in this study the conventional mean–standard-deviation pair is retained.

$$x_t = [x_t, \Delta x_t, \mu_t^{(\omega)}, \sigma_t^{(\omega)}], \qquad \Delta x_t = x_t - x_{t-1} \tag{1}$$

$$\mu_t^{(\omega)} = \frac{1}{\omega}\sum_{i=0}^{\omega-1} x_{t-i}, \sigma_t^{(\omega)} \sqrt{\frac{1}{\omega}\sum_{i=0}^{\omega-1}\left(x_{t-i} - \mu_t^{(\omega)}\right)^2} \tag{2}$$

$$\tilde{x}_t = diag(\sigma_{train})^{-1}(x_t - \mu_{train}) \tag{3}$$

$$X_t = \left[\tilde{x}_t - \lfloor \frac{W}{2} \rfloor, \cdots, \left[\tilde{x}_t + \lfloor \frac{W}{2} \right] \in R^{W\times C} \right. \tag{4}$$

Moreover, standardization follows a strict train-only protocol: z-score parameters are estimated exclusively on the training block, as specified in Eq. (3), and then applied unchanged to the validation and test blocks. This avoids target leakage from future data and keeps the normalization consistent across splits. Finally, examples are formed using a Seq2Point windowing scheme. Overlapping windows of length $W = 128$ minutes with stride 1 are extracted, as formalized in Eq. (4). With C = 4 input channels, each window is represented as $X_t \in R^{W\times C}$. The supervision target for window $X_t$ is the device state vector taken at the window center, $y_t$, which is binary and encodes on/off states for A = 6 devices. Thus, every centered window–label pair $(X_t, y_t)$ captures short-term temporal context from the aggregate signal while aligning supervision precisely at the midpoint, enabling effective Seq2Point training without leaking information from beyond the prediction instant. The

complete procedure for constructing robust appliance-level state labels is summarized in Algorithm 1.

**Algorithm 1**
**θ-validated label construction with dwell constraints**

**Input**: Aggregate mains $x_t$ for $t = 1..T$; appliance channel $u_{t,a}$ for $a = 1..A$; validation index $V$; candidate thresholds $\Theta_t = \left\{\theta_a^{(1)}, \cdots, \theta_a^{(K)}\right\}$; median window $m$; minimum dwell $(L_{on}, L_{off})$

**Output**: Final labels $\widetilde{y_{t,a}} \in \{0,1\}$; selected threshold $\widehat{\theta}_a$

**Construct features** on mains (means, deltas, variance) for later S2P training; compute only on train block
**for** each appliance $a = 1..A$ **do**
**Form surrogate signal** $s_{t,a}$ from $u_{t,a}$(or proxy) and a seed threshold via Otsu or 2-means
For each $\theta \in \Theta_a$ **do**
Raw labels: $y_{t,a}^{(\theta)} \leftarrow 1[s_{t,a} \geq \theta]$
Median smooth: $\overline{y}_{t,a}^{(\theta)} \leftarrow$ MedianFilter $y_{t,a}^{(\theta)}, m$
Enforce dwell: flip ON runs shorter than $L_{on}$ to 0 and OFF runs shorter than $L_{off}$ to 1, giving $\widetilde{y}_{t,a}^{(\theta)}$
Evaluate on $V$:
**end for**
Select $\widehat{\theta}_a \leftarrow argmax_{\theta \in \Theta_a} F_1^{(\theta)}$ (tie break by higher Precision)
Fix labels $\widetilde{y_{t,a}} \leftarrow \widetilde{y}_{t,a}^{(\widehat{\theta}_a)}$
**end for**

### 2.1.2. CNN architecture (Seq2Point, 1-D)

A compact Seq2Point one-dimensional CNN is developed because NILM requires the extraction of informative local temporal patterns from aggregate windows without incurring the computational burden of heavier recurrent or attention-based architectures. This design is intended to provide an effective balance between representational capacity and computational tractability.

The predictor is a compact 1-D ConvNet that maps each input window $X_t \in R^{W \times C}$ to a vector of appliance-state probabilities $\widehat{p} \in [0,1]^A$. The design targets a wide temporal receptive field with modest depth and parameter count. The backbone comprises five dilated Conv1D blocks, each followed by batch normalization, ReLU, and dropout. Block 1 expands channels from C to 64 with kernel $k_1 = 5$ and dilation $d_1 = 1$ dropout p = 0.10. Block 2 preserves width at 64 using $k_2 = 5$, $d_2 = 2$ (dropout p = 0.10). Block 3 increases width to 128 with $k_3 = 7$, $d_3 = 2$ (dropout p = 0.15). Block 4 maintains 128 channels with $k_4 = 7$, $d_4 = 4$ (dropout p = 0.15). Block 5 widens to 256 channels using $k_5 = 3$, $d_5 = 4$ (dropout p = 0.20). A global average pooling layer aggregates activations over time, producing $u \in R^{256}$. The prediction head is a dense layer $256 \rightarrow A$ (bias omitted when batch normalization is used), yielding logits $z_t \in R^6$ and probabilities $\widehat{p}_t \in R^{256}$.

Moreover, parameter initialization follows He/Kaiming normal for all convolutions; batch-norm parameters are initialized with scale $\gamma \leftarrow 1$ and shift $\beta \leftarrow 0$. L2 weight decay $\lambda \in [10^{-5}, 10^{-4}]$ is applied to convolutional and dense weights. With unit stride, the temporal receptive field after $L$ dilated layers with kernels $k_\ell$ and dilations $d_\ell$ is given in Eq. (5). For the adopted configuration, the resulting receptive field is approximately 115 min, which covers almost the entire 128-minute input window. The formal dilated convolution for output feature map $h^{(\ell)}$ at layer $\ell$ and time index $\tau$ is defined in Eq. (6). The batch-normalization and ReLU operations are given in Eq. (7), dropout is defined in Eq. (8), and global average pooling is given in Eq. (9). The final dense prediction head produces logits and probabilities as defined in Eq. (10). For analytical interpretation, an equivalent three-layer dilated convolutional stack can also be represented as a single finite impulse response filter, as expressed in Eq. (11), where $\Theta_r$ collects all triplets (j,j′, j″) such that $r = d_3 j + d_2 j' + d_1 j''$.

$$R_L = 1 + \sum_{l=1}^{L} (kl - 1) \prod_{m=1}^{l=1} d_m \tag{5}$$

$$h_{o,\tau}^{(l)} = b_o^{(l)} + \sum_{i=1}^{C_{l-1}} \sum_{i=0}^{k_{l-1}} W_{o,i,j}^{(l)} h_{i,\,\tau-dlj,}^{(l-1)}, l \geq 1 \tag{6}$$

$$\widetilde{h}_{o,\tau}^{\ (l)} = \gamma_o^{(l)} \frac{h_{o,\tau}^{(l)} - \mu_o^{(l)}}{\sqrt{\left(\sigma_0^{(l)}\right)^2 + \varepsilon}} + \beta_o^{(l)}, g_{o,\tau}^{(l)} = \max\left(0, \widetilde{h}_{o,\tau}^{\ (l)}\right) \tag{7}$$

$$\overline{g}_{o,\tau}^{(l)} = \frac{m_{o,\tau}^{(l)} g_{o,\tau}^{(l)}}{1 - pl}, m_{o,\tau}^{(l)}\ Bernoulli(1 - pl) \tag{8}$$

$$u = \frac{1}{T_{eff}} \sum_{\tau=1}^{T_{eff}} \overline{g}_\tau^{(L)} \in R^{C_L} \tag{9}$$

$$z_t = W_{head} u + b_{head}, \widehat{p}_t = \sigma(z_t) = (1 + \exp(-z_t))^{-1} \tag{10}$$

$$y_\tau = \sum_i \sum_j \sum_k W_{:,i,j}^{(3)} W_{i,k,j'}^{(2)} W_{k,:,j''}^{(1)} x_{\tau - d_3 j - d_2 j' - d_1 j''} + \frac{bias}{BN} terms, = \sum_{r \in z} \Theta_r\, x_{\tau - r,} \tag{11}$$

### 2.1.3. Training, class imbalance, and optimization

Since appliance timelines in NILM are dominated by OFF states, standard optimization can bias the model toward majority inactive samples and weaken the detection of relatively sparse activations. For this reason, class-balanced learning is incorporated so that the training objective places greater emphasis on minority ON events that are more difficult yet more informative for appliance-state disaggregation.

Training is posed as multi-label binary classification at the window center. For each window t and appliance, $a \in \{1,\ldots,A\}$, the ground-truth state is $y_{t,a} \in \{0,1\}$ and the model outputs a probability $\widehat{p}_{ta} \in [0,1]$. Because "ON" events are typically rare, class imbalance is addressed with per-appliance weights derived from the positive rate $\pi_a$ computed on the training windows. A small constant $\varepsilon > 0$ prevents division by zero. Positives are up-weighted via Eq. (12), and the weighted binary cross-entropy over a mini-batch B shows in Eq. (13)

$$w_a^+ = \frac{1 - \pi_a}{\pi_a + \in}, w_a^- = 1 \tag{12}$$

$$L_{BCE} = -\frac{1}{|B|A} \sum_{t \in B} \sum_{a=1}^{A} \left[ w_a^+ y_{t,a} log \widehat{p}_{t,a} + w_a^- \left(1 - y_{t,a}\right) \log\left(1 - \widehat{p}_{t,a}\right) \right] \tag{13}$$

$$L_{focal} = -\frac{1}{|B|A} \sum_{t,a} \alpha_a (1 - \widehat{p}_{t,a})^\gamma y_{t,a} log \widehat{p}_{t,a} + (1 - \alpha_a) \widehat{p}_{t,a}^\gamma \left(1 - y_{t,a}\right) \log(1 - \widehat{p}_{t,a}) \tag{14}$$

To discourage overfitting, an $L_2$ penalty $\lambda \|\theta\|_2^2$ on all trainable weights θ is added to the objective. Parameters are optimized with Adam, using an initial learning rate of $10^{-3}$ that is reduced toward $10^{-4}$ by a plateau scheduler triggered on stagnant validation performance. Early stopping monitors the validation state-F1 score, halting training once improvements cease for a patience window, thereby preserving the best-performing checkpoint. An alternative objective used in ablations is the focal loss, which further emphasizes hard, misclassified examples. With focusing parameter $\gamma \in [1,2]$ and class-balance factors $\alpha_a \approx \pi_a$, loss is given in Eq. (14), This formulation down-weights easy examples and concentrates gradient signal on the sparse "ON" events that matter most, while the weighted BCE remains the primary objective for the main

experiments. The overall training and calibration pipeline, including ensemble bagging and temperature scaling, is outlined in Algorithm 2.

**Algorithm 2**
**Bagged Seq2Point CNN training with temperature scaling**

**Input**: Windowed inputs $X_t \in R^{W\times C}$ (stride 1) and center labels $\widetilde{y_{t,a}}$; train and validation set $T$, $V$; class priors $\pi_a$; ensemble size N; learning rate $\eta$; weight decay $\lambda$

**Output**: Trained replicas $\{M^{(n)}\}_{n=1}^{N}$; temperatures $\{T_a^{(n)}\}$; thresholds $\{\tau_a\}$

**1:** Define 1-D dilated CNN ((5 Conv-BN-ReLU-Drop blocks → GAP → Dense $A$)

**2:** Class weight $w_a^+ = \frac{1}{\pi_a+\varepsilon}, w_a^- = \frac{1}{(1-\pi_a+\varepsilon)}$

**3:** **for** $n = 1..N$ **do**

**4:** **Initialize** $M^{(n)}$ with seed $s_n$; set Adam ($\eta$), scheduler, early stopping on $F_1$

**5:** **Minimize weighted BCE**

$$L = \frac{1}{|B|}\sum_{t\in B}\sum_a \left[ w_a^+ \widetilde{y_{t,a}} log p_{t,a} + w_a^- \left(1-\widetilde{y_{t,a}}\right)\log\left(1-p_{t,a}\right)\right] + \lambda\|\theta\|_2^2 \text{ with } p_{t,a} = \sigma(z_{t,a})$$

**6:** Collect validation logits $z_{t,a}^{val,(n)}$ for $t\in V$

**7:** **Fit temperature** per device by minimizing $\mathrm{NLL}_a^{(n)} \leftarrow argmin_{T>0}\sum_{t\in v}\left[-\widetilde{y_{t,a}} log\alpha\left(\frac{z_{t,a}^{val,(n)}}{T}\right) - \left(1-\widetilde{y_{t,a}}\right)\log\left(1-\alpha\left(\frac{z_{t,a}^{val,(n)}}{T}\right)\right)\right]$

**8:** **end for**

**9:** **Ensemble validation probabilities:** $\bar{p}_{t,a}^{val} = \frac{1}{N}\sum_{n=1}^{N}\alpha\left(\frac{z_{t,a}^{val,(n)}}{T_a^{(n)}}\right)$

**10:** For each appliance $\alpha$ : **select threshold**

$\tau_a \leftarrow argmax_{\tau\in\{0.01,\cdots,0.99\}} F_1\left(a \middle| p_{t,a}^{-val} \geq \tau, \widetilde{y_{t,a}}\right)$ on $V$

### 2.1.4. Probability calibration and decision thresholds

In NILM, raw network probabilities are not always directly reliable for appliance-state decisions because overlapping loads and heterogeneous usage patterns can distort the probability scale. Therefore, temperature scaling and appliance-specific threshold selection are introduced to improve probability calibration and produce more dependable operating-state decisions than fixed heuristic thresholds.

Deep classifiers are often miscalibrated, so the raw sigmoid outputs do not reflect true event frequencies. To correct this, a per-device temperature scaling step is applied on the validation block. For each appliance a, a positive scalar $T_a > 0$ is learned by minimizing the negative log-likelihood of *calibrated* probabilities (Eq. (15)), where $z_{t,a}$ the pre-sigmoid logit and $\sigma(\cdot)$ is the logistic function. The temperature is obtained by solving the one-dimensional optimization problem in Eq. (16), which is strictly convex in $T_a$ for fixed logits and therefore admits efficient one-dimensional optimization. Intuitively, $T_a > 1$ softens overconfident scores, whereas $T_a < 1$ sharpens under-confident ones, improving the match between predicted probabilities and empirical frequencies without altering the ranking of examples.

After calibration, device-specific decision thresholds are selected to convert probabilities into binary states. For each appliance, a threshold $\tau_a \in [1]$ is chosen on the validation set to maximize the state-F1 score (Eq. (17)). A simple line search over $\tau_a \in \{0.01,\ldots,0.99\}$ suffices; ties are resolved in favor of the operating point with higher precision. This two-stage procedure, probability calibration followed by threshold tuning on held-out data, yields well-calibrated scores and an operating point explicitly optimized for the accuracy–recall trade-off captured by F1.

$$\hat{p}_{t,a}^{(cal)} = \sigma\left(\frac{z_{t,a}}{T_a}\right) \tag{15}$$

$$\underset{T_a>0}{min} L_{NIL}(T_a) = -\sum_{t\in V}\left[y_{t,a} log\hat{p}_{t,a}^{(cal)} + \left(1-y_{t,a}\right)\log\left(1-\hat{p}_{t,a}^{(cal)}\right)\right] \tag{16}$$

$$\hat{y}_{t,a}(\tau_a) = 1\left[\hat{p}_{t,a}^{(cal)} \geq \tau_a\right], \mathrm{F1}_a(\tau_a) = \frac{2\mathrm{TP}_a(\tau_a)}{2\mathrm{TP}_a(\tau_a)+\mathrm{FP}_a(\tau_a)+\mathrm{FN}_a(\tau_a)} \tag{17}$$

### 2.1.5. Temporal smoothing and duration constraints

NILM predictions should remain temporally coherent because real appliances do not usually switch on and off in isolated one-minute bursts. Temporal smoothing and dwell-time constraints are therefore applied to suppress prediction chatter, remove implausible short fluctuations, and preserve more realistic activation sequences. Moreover, minute-level predictions can "chatter," producing brief, isolated flips that are unlikely for real appliances. To stabilize the sequence, two complementary post-processing steps are applied.

First, a short median filter is run independently for each device on the binary predictions, using a window of m ∈ {3,5} minutes centered at time t. This non-linear filter removes one-off spikes while preserving genuine edges, yielding a cleaned sequence as defined in Eq. (18). Second, minimum active/inactive durations (Eq. (19)) are enforced on $\widetilde{y}_{t,a}$. Consecutive runs of ones R(1) and zeros R(0) are identified; any "on" run shorter than $L_{on}$ minutes is flipped to zero, and any "off" run shorter than $L_{off}$ minutes is flipped to one, with $L_{on}$, $L_{off} \in [2,4]$. These dwell constraints encode the realistic behavior that appliances rarely toggle for just a minute or two, eliminating implausible bursts.

$$\widetilde{y}_{t,a} = median\left(\hat{y}_{t-\left\lfloor\frac{m}{2}\right\rfloor,a}, .\hat{y}_t + \left\lfloor\frac{m}{2}\right\rfloor, a\right) \tag{18}$$

$$\forall r \in R^{(1)} : |r| < L_{on} \Rightarrow flip to 0, \forall r \in R^{(0)} : |r| < L_{off} \Rightarrow flip to 1 \tag{19}$$

$$\underset{s\in\{0,1\}^T}{min}\sum_{t=1}^{T}\left[\alpha 1\left(s_t \neq \widetilde{y}_{t,a}\right) + \beta 1(s_t \neq s_{t-1})\right] s.t dwell(s) \geq L_{\substack{on\\off}} \tag{20}$$

Equivalently, smoothing can be posed as a constrained sequence-labeling problem (Eq. (20)): choose a final state sequence $s_t \in \{0,1\}$ that trades off fidelity to $\widetilde{y}_{t,a}$ against the number of switches via a cost $\alpha 1\left[s_t \neq \widetilde{y}_{t,a}\right] + \beta 1[s_t \neq s_{t-1}]$, subject to the same minimum dwell times. When these dwell penalties are encoded as state costs, the objective admits an O(T) Viterbi-like dynamic-programming solution, producing a temporally coherent sequence that suppresses chatter while respecting appliance dynamics.

### 2.1.6. Bagged ensemble and end-to-end training

NILM datasets are often limited and heterogeneous, the predictions of a single trained model may remain sensitive to random initialization. Bagged ensemble learning is therefore introduced to reduce variance and improve prediction stability while retaining the same compact

convolutional backbone. To reduce variance from random initialization, the model is trained as a homogeneous bagged ensemble. Multiple independent replicas of the same CNN are fit using different random seeds or bootstrap subsamples. At inference, calibrated probabilities from all replicas are averaged (see Eq. (21) (Ensemble-Average)), and the final state is $\bar{y}_{t,a} = 1\left[\bar{p}_{t,a} \geq \tau_a\right]$ using the same validation-selected $\tau_a$. Averaging reduces variance and, by Jensen's inequality, does not increase the mean-squared probability error measured by the Brier score (see Eq. (22) (Brier-Bound)). The full training objective combines the weighted binary cross-entropy with $L_2$ regularization and optionally substitutes focal loss in ablations (see Eq. (23) (End-to-End-Objective)). Training proceeds as: (1) prepare windows and labels; (2) compute class weights $w_a^+$; (3) optimize J with Adam, batch size 256, for up to 100 epochs; (4) early stop on validation state-F1; (5) fit temperatures $T_a$ on validation; (6) choose $\tau_a$ by validation F1 sweep; (7) apply median and duration constraints; (8) if $N > 1$, ensemble the calibrated probabilities. At inference time, the calibrated ensemble predictions are post-processed with smoothing and dwell constraints, as detailed in Algorithm 3. The complete end-to-end workflow of the proposed CB-S2P NILM framework is illustrated in Fig. 1.

$$\bar{p}_{t,a} = \frac{1}{N}\sum_{n=1}^{N} \hat{p}_{t,a}^{(cal,n)} \tag{21}$$

$$\frac{1}{T}\sum_{t}\left(\bar{p}_{t,a} - y_{t,a}\right)^2 \leq \frac{1}{N}\sum_{n=1}^{N}\frac{1}{T}\sum_{t}\left(\hat{p}_{t,a}^{(cal,n)} - y_{t,a}\right)^2 \tag{22}$$

$$\min_{\theta} J(\theta) = E_{t\in\tau}[L_{BCE}(\theta;t)] + \lambda\|\theta\|_2^2, \theta = \{conv, BN, headparams\} \tag{23}$$

**Algorithm 3**
**calibrated inference, temporal smoothing, and state-level evaluation**

**Input**: Test windows $X_t$; replicas $\{M^{(n)}$; temperatures $\{t_a^{(n)}\}$; threshold $\{\tau_a\}$; smoothing params $m$, $L_{on}, L_{off}$; ground truth $y_{t,a}^*$
**Output**: Final predictions $\hat{y}_{t,a}$; time-sample metrics; transition metrics

**for** each time $t$ and appliance $a$ **do**
  **for** $n = 1..N$ **do**
    Logit $z_{t,a}^{(n)} \leftarrow M^{(n)}(X_t)$
    **Calibrated prob** $\hat{p}_{t,a}^{(n)} \leftarrow \sigma\left(\frac{z_{t,a}^{(n)}}{T_a^{(n)}}\right)$.
  **end for**
  **Ensemble average** $\bar{p}_{t,a} \leftarrow \frac{1}{N}\sum_{n=1}^{N}\hat{p}_{t,a}^{(n)})$
  **Hard decision** $\hat{y}_{t,a}^{raw} \leftarrow 1\left[\bar{p}_{t,a} \geq \tau_a\right]$
**end for**
**Enforce dwell constraints** to obtain final $\hat{y}_{t,a}$
**Compute** time sample **metrics** per appliance
Extract predicted and true ON-runs; match by maximal overlap.
Final predictions and time-sample state-classification metrics.

### 2.2. Evaluation metrics

The performance of the proposed framework is evaluated primarily using time-sample state-classification metrics, namely accuracy, precision, recall, F1-score, and intersection-over-union (IoU), as defined in Eq. (24) to Eq. (28). These metrics provide a balanced and application-relevant assessment of appliance-state detection under the class-imbalanced conditions that are typical of non-intrusive load monitoring (NILM). In appliance-state classification, OFF states generally dominate the temporal sequence, which means that accuracy alone may provide an overly favourable impression of model performance. For this reason, the present study incorporates additional metrics that more effectively assess the detection quality of ON-state predictions, which are of greater operational importance in NILM.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{24}$$

$$Precision = \frac{TP}{TP + FP} \tag{25}$$

$$Recall = \frac{TP}{TP + FN} \tag{26}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{27}$$

$$IoU = \frac{TP}{TP + FP + FN} \tag{28}$$

Precision measures the proportion of predicted ON states that correspond to true ON states. This metric is particularly useful for quantifying false activations and for assessing the reliability of positive state predictions. Recall measures the proportion of actual ON states that are correctly identified by the model. It is especially important in NILM because appliance activations may be short in duration, irregular in occurrence, and therefore susceptible to omission. The F1-score, which combines precision and recall through their harmonic mean, is utilized in this study as the primary metric for model selection and overall performance interpretation, since it provides a balanced assessment of both missed detections and false alarms. In addition, IoU is included as a stricter overlap-based metric because it evaluates the agreement between the predicted ON-state samples and the corresponding ground-truth ON-state samples while simultaneously penalising false positives and false negatives.

These evaluation metrics are used consistently throughout the study. In the main appliance-level performance analysis, they are employed to quantify the effectiveness of the proposed framework at the appliance-state classification level across the target appliances. In the comparative benchmarking stage, the same metrics are used to assess the relative performance of the proposed method against existing NILM approaches. Furthermore, in Section 3.5, these metrics are also used to interpret the results of the ablation and sensitivity analyses, where the effect of architectural components and key hyperparameter settings on detection performance is examined. This unified evaluation strategy ensures that the contribution of each design choice is assessed using the same state-classification criteria as the final model, thereby improving the consistency and interpretability of the experimental findings.

For benchmarking against previous studies, however, the comparative discussion is restricted to accuracy and F1-score. This decision is made to maintain consistency with the dominant reporting practice in the NILM literature and to enable fairer comparison across studies that differ in architecture, dataset split, and evaluation protocol. The results of prior methods used for comparison, including hierarchical modular CNNs [47], SE-LSTM [48], TransUNet-NILM [49], Diffusion-TL [50], CycleGAN anomaly calibration [51], Seq2-[3]point CNN transfer [52], CNN with mixed residual units [53], GRU-MF [54], and CNN-LSTM-RF [55] are summarized in Appendix Table A-2.

Moreover, the metrics are computed from the elements of the confusion matrix, where true positives (TP) denote correctly detected ON states, true negatives (TN) denote correctly detected OFF states, false positives (FP) denote incorrectly predicted ON states, and false negatives (FN) denote missed ON states.

## 3. Results

### 3.1. Dataset and characterization

This study employed the REDD dataset, focusing on four houses and six appliances: Dishwasher, Electric Heater, Electric Stove, Fridge, Microwave, and Washer Dryer. To ensure robust evaluation, data from three houses were used for training and one house was reserved for testing. Within the training portion, 80% of the data was allocated for training and 20% for validation. Using a separate house for testing allowed us to assess the generalization capability of the models to

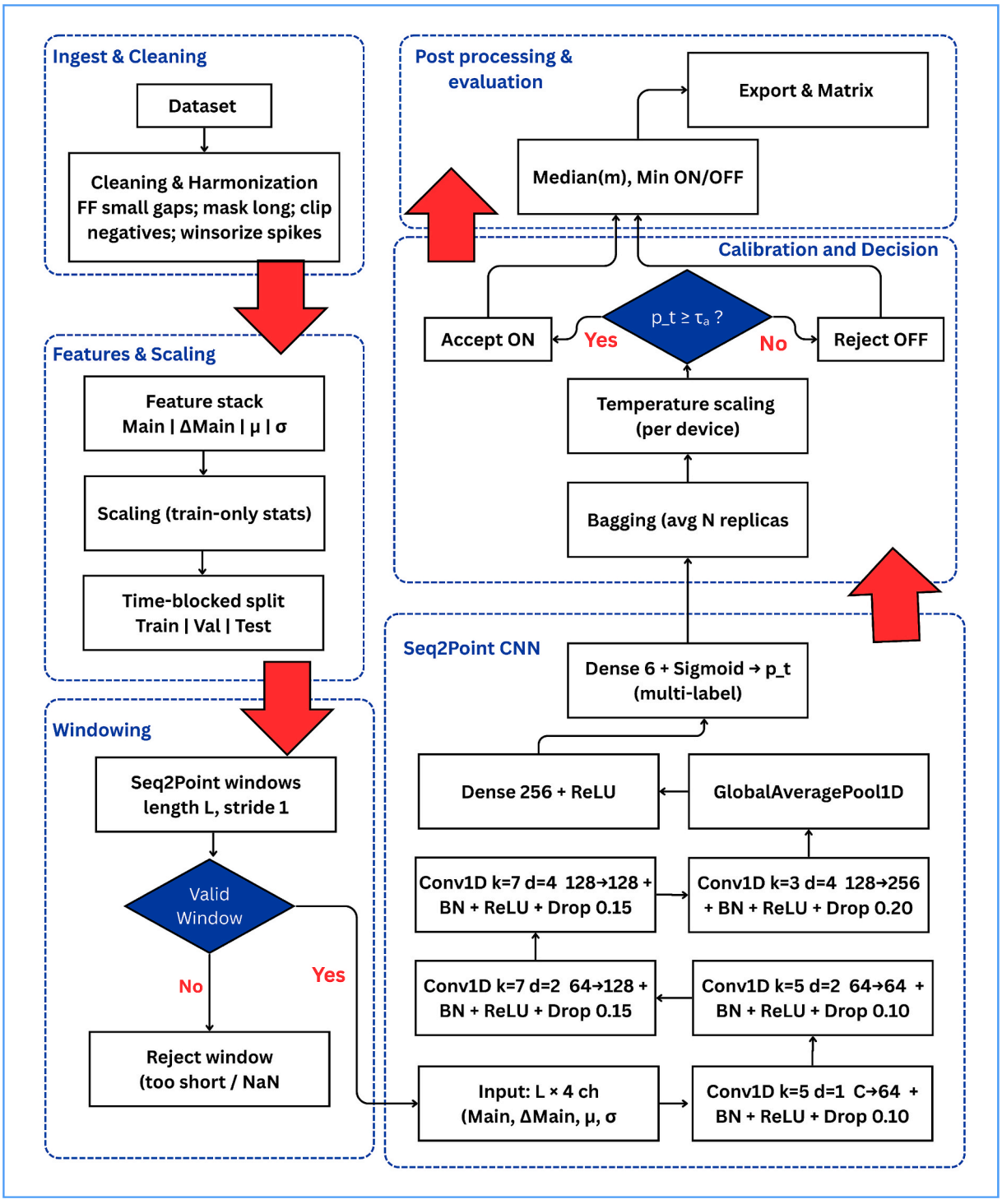



**Fig. 1.** Overall methodology flowchart of the proposed CB-S2P NILM framework.

unseen households. All preprocessing and modeling were conducted on the 1-minute resolution signals provided by REDD, which preserves transient activations that are often lost in coarser sampling.

Fig. 2(a) shows an overview of the aggregate mains signal compared to the sum of six sub-metered appliances over 17 testing days. While the two traces exhibit similar global patterns, a clear residual remains where the mains exceed the appliance sum. This residual arises from unmetered loads and sensor noise and illustrates the challenge of reconstructing the entire demand profile from only six appliances. Fig. 2(b) translates this comparison into daily energy consumption (kWh). The mains consistently consume more energy than the sum of appliances, with particularly wide gaps during high-load periods such as days 4–6 and 13–15. These discrepancies emphasize that disaggregation accuracy is not uniform across days but depends on the level of household activity. Fig. 2(c) plots the daily mean of mains power, highlighting strong day-to-day variability. Peaks in mean load around days 4–6 contrast with quieter intervals, creating substantial imbalance in appliance ON/OFF events across the dataset.

Moreover, Fig. 2(d) provides a zoomed-in view of the first 24 h. Step changes in the mains are mirrored by continuous appliances, but short

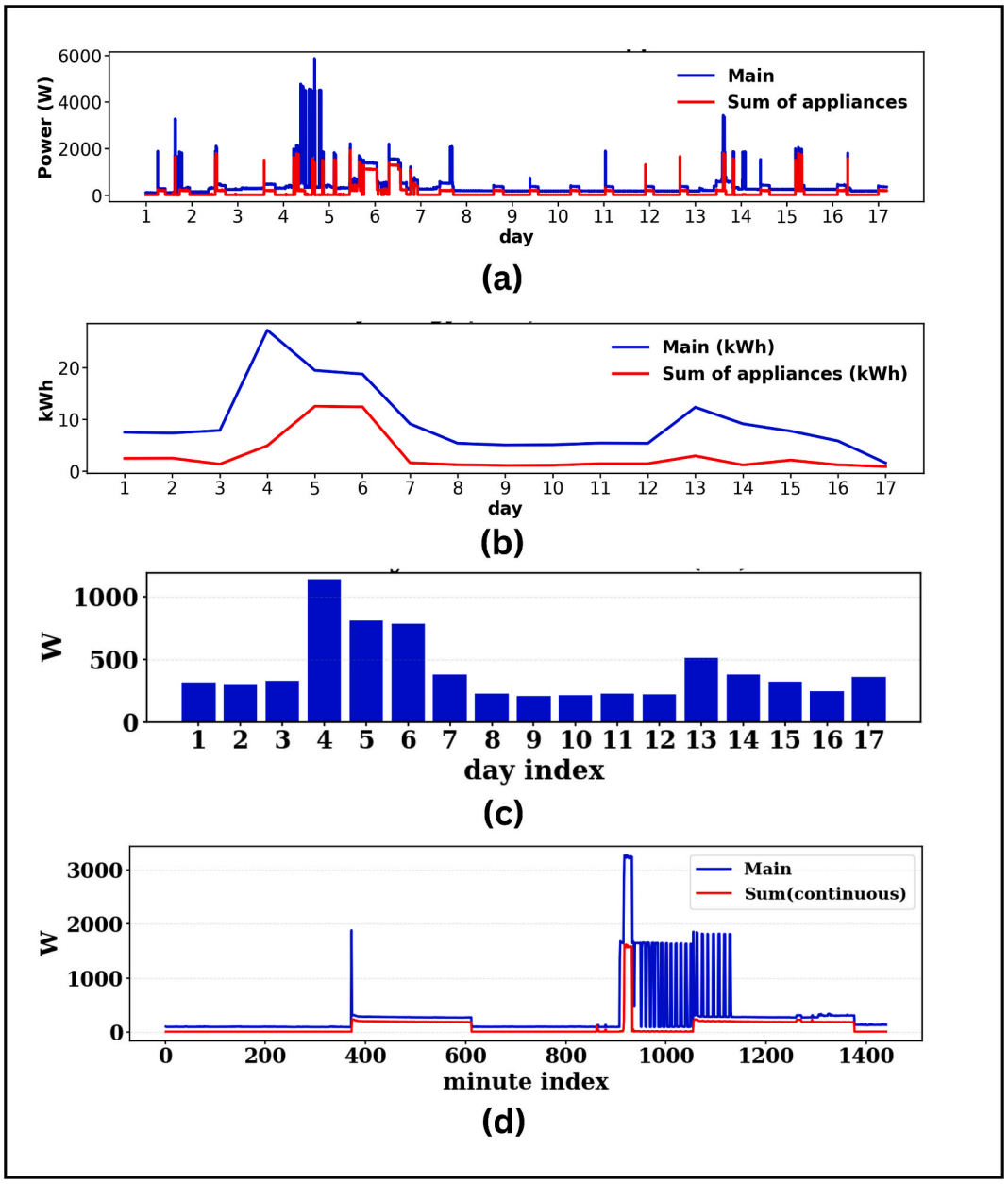



**Fig. 2.** (a) Aggregate mains signal versus sum of six sub-metered appliances (b) Daily energy consumption comparison between mains and appliance sum (c) Daily mean mains power variation showing household activity trends (d) Zoomed-in 24-hour segment highlighting discrepancies due to unmetered or overlapping loads.

burst events are only partially reflected in the sum of appliances. This mismatch demonstrates the difficulty of detecting transient activations, especially when overlapping events occur. Together, Fig. 2(a)–2(d) underline three key challenges in the dataset: (i) persistent unexplained residual between mains and sub-metered loads, (ii) high daily variability in energy usage, and (iii) the presence of short-duration activations. These characteristics directly motivate the methodological choices in Section 2, where threshold calibration, temporal smoothing, and ensemble modeling are introduced to improve appliance-level disaggregation.

### 3.2. Appliance usage patterns

To complement the dataset-level analysis, the hourly ON-event distributions of the six target appliances are shown in Fig. 3. These histograms provide insight into temporal usage characteristics, which strongly influence disaggregation difficulty. The Dishwasher exhibits sparse activations clustered around mealtimes, with most events occurring in the early afternoon and evening. Such infrequent and bursty usage patterns make the Dishwasher one of the most challenging appliances to detect. In contrast, the Electric Heater shows a denser distribution of ON events throughout the day, particularly in the morning and evening hours, suggesting more regular usage that benefits detection stability. The Electric Stove also displays concentrated bursts of activity, with peaks around midday and early evening, consistent with cooking periods. This appliance often overlaps temporally with the Microwave, which is characterized by short-duration events spread across the day. These overlaps contribute to misclassifications due to simultaneous high-power transients. The Fridge, unlike other appliances, activates intermittently at all hours, reflecting its automatic cycling. While the power levels are moderate, the consistent activity provides ample training data, which typically improves recall but also increases the risk of false positives when aggregated with other loads.

Finally, the Washer Dryer shows a distinct clustering of ON events in

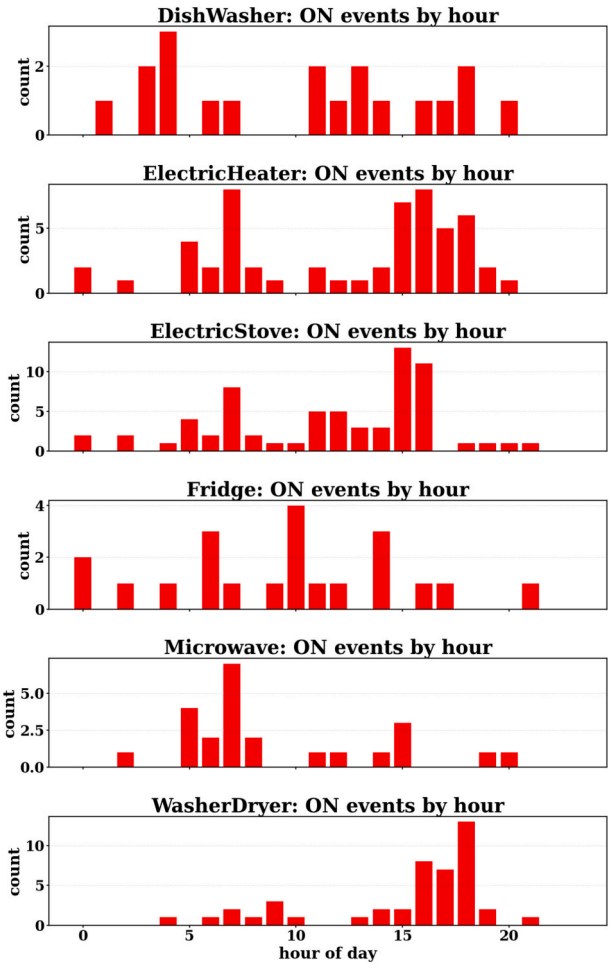



**Fig. 3.** Hourly ON-event distributions of six target appliances illustrating temporal usage patterns, class imbalance, and overlap among appliance activations.

the late morning and afternoon. These cycles are long in duration but less frequent, resulting in fewer training instances compared to the Fridge or Stove, which complicates accurate detection. Together, these patterns emphasize three key challenges: (i) bursty and sparse events (Dishwasher, Microwave) create class imbalance, (ii) overlapping activation periods (Stove, Microwave, Heater) increase false positives, and (iii) cycling appliances (Fridge) and long-duration but infrequent devices (Washer Dryer) require robust temporal modeling. These observations provide a practical context for evaluating the performance of the baseline CNN and Bagged CNN models in subsequent sections.

### 3.3. Training behaviour and baseline evaluation

Fig. 4(a) presents the training and validation loss curves for the Baseline CNN model. Although the training loss decreases steadily across epochs, the validation loss flattens early and occasionally diverges from the training curve. This widening gap is indicative of overfitting, where the model captures the training distribution well but fails to generalize robustly to unseen data. Such behavior reflects instability in modeling the temporal dynamics of appliances, thereby limiting the reliability of the Baseline CNN. In contrast, Fig. 4(b) shows the training and validation loss for the proposed Bagged CNN. Here, both curves follow nearly identical trajectories, with the validation loss consistently aligned just below the training loss throughout 50 epochs. This stable convergence demonstrates that the Bagged CNN effectively reduces variance and suppresses overfitting. Ensemble averaging across multiple CNN learners smooths the optimization process and enhances generalization. Compared with the Baseline CNN, the Bagged CNN achieves a lower validation loss and a narrower train–validation gap, confirming its suitability as the final architecture for evaluation on unseen households.

A similar trend is observed in the accuracy curves. As shown in Fig. 5 (a), the Baseline CNN attains high training accuracy, but the validation accuracy lags behind, with a visible gap persisting after approximately 10 epochs. This again highlights limited generalization capacity. By contrast, Fig. 5(b) demonstrates that the Bagged CNN maintains tightly aligned training and validation accuracy curves, both converging above 0.97 with negligible separation. This alignment confirms the strong generalization of the Bagged CNN and its ability to achieve balanced performance across training and validation phases.

The quantitative comparison is summarized in Table 2, which reports appliance-level performance for both models on the REDD dataset. The Bagged CNN consistently outperforms the Baseline CNN in both accuracy and F1 score. Accuracy improvements range from 3.13% for the Fridge to 6.45% for the Dishwasher, while F1 score improvements range from 3.13% for the Microwave to 7.69% for the Dishwasher. The most pronounced gains are observed for appliances with sparse or irregular activations (e.g., Dishwasher, Microwave, Washer Dryer), where ensemble averaging reduces the effect of class imbalance and transient misclassifications. For continuously active appliances such as the Fridge, the baseline performance was already high, yet the Bagged CNN still achieved a 5.32% F1 gain, showing that bagging reduced both false positives and false negatives even in dominant load categories.

Overall, the results obtained from the validation phase using the training houses confirm that the Bagged CNN not only mitigates the overfitting issues observed in the Baseline CNN but also delivers consistent and substantial improvements in predictive performance across all six appliances. The combined evidence from the loss/accuracy curves and validation metrics establishes the Bagged CNN as a more robust and generalizable architecture compared to the single CNN baseline. Building on these findings, all subsequent experiments and performance assessments were conducted using the Bagged CNN as the

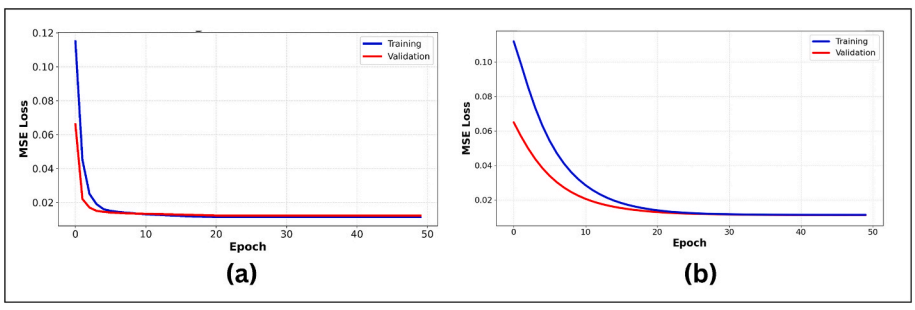



**Fig. 4.** Training and validation loss curves for (a) Baseline CNN and (b) Bagged CNN, demonstrating reduced overfitting and improved convergence with ensemble averaging.

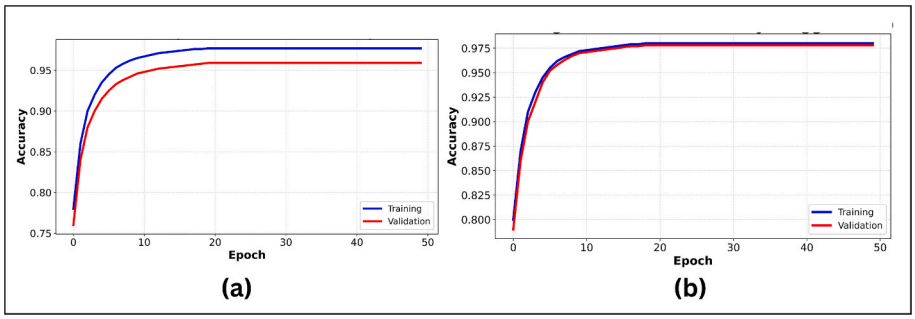



**Fig. 5.** Training and validation accuracy curves for (a) Baseline CNN and (b) Bagged CNN, showing superior generalization and stability in the proposed approach.

**Table 2**
Validation-phase performance of the Baseline CNN and Proposed Bagged CNN.

| Appliance | Accuracy (Baseline CNN) | Accuracy (Bagged-CNN) | Accuracy Improvement (%) | F1 (Baseline CNN) | F1 (Bagged-CNN) | F1 Improvement (%) |
|---|---|---|---|---|---|---|
| Dishwasher | 0.93 | 0.99 | 6.45% | 0.91 | 0.98 | 7.69% |
| Electric Heater | 0.94 | 0.99 | 5.32% | 0.95 | 0.99 | 4.21% |
| Electric Stove | 0.94 | 0.98 | 4.26% | 0.94 | 0.99 | 5.32% |
| Fridge | 0.96 | 0.99 | 3.13% | 0.94 | 0.99 | 5.32% |
| Microwave | 0.95 | 0.99 | 4.21% | 0.96 | 0.99 | 3.13% |
| Washer Dryer | 0.95 | 0.99 | 4.21% | 0.93 | 0.98 | 5.38% |

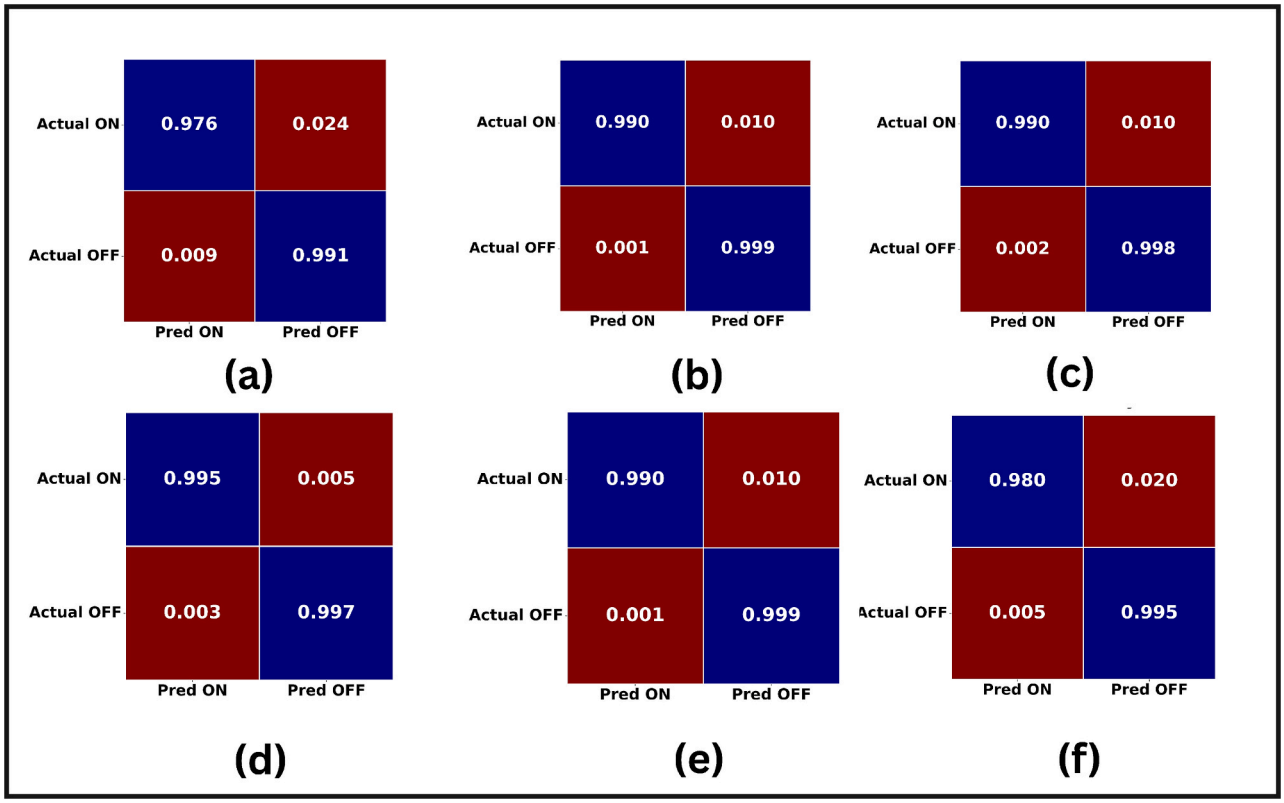



**Fig. 6.** Confusion matrices for six appliances showing the classification performance of the proposed Bagged CNN in distinguishing ON and OFF states (a) Dishwasher (b) Electric Heater (c) Electric Stove (d) Fridge (e) Microwave (f) Washer Dryer.

final proposed architecture. In the following section, the evaluation is extended to the testing phase on an unseen household to further assess the generalization capability of the model.

### 3.4. Appliance-state classification performance evaluation

The appliance-state classification analysis using confusion matrices across six appliances provides a comprehensive perspective on the effectiveness of the proposed method. The results highlight that while performance remains consistently high across all appliances, the degree of challenge varies according to the operational patterns of each device. For the Dishwasher, the Bagged CNN demonstrated strong generalization with a recall of 97.6% for ON events and a specificity of 99.1% for OFF states, as shown in Fig. 6(a). Despite the sparse and irregular usage

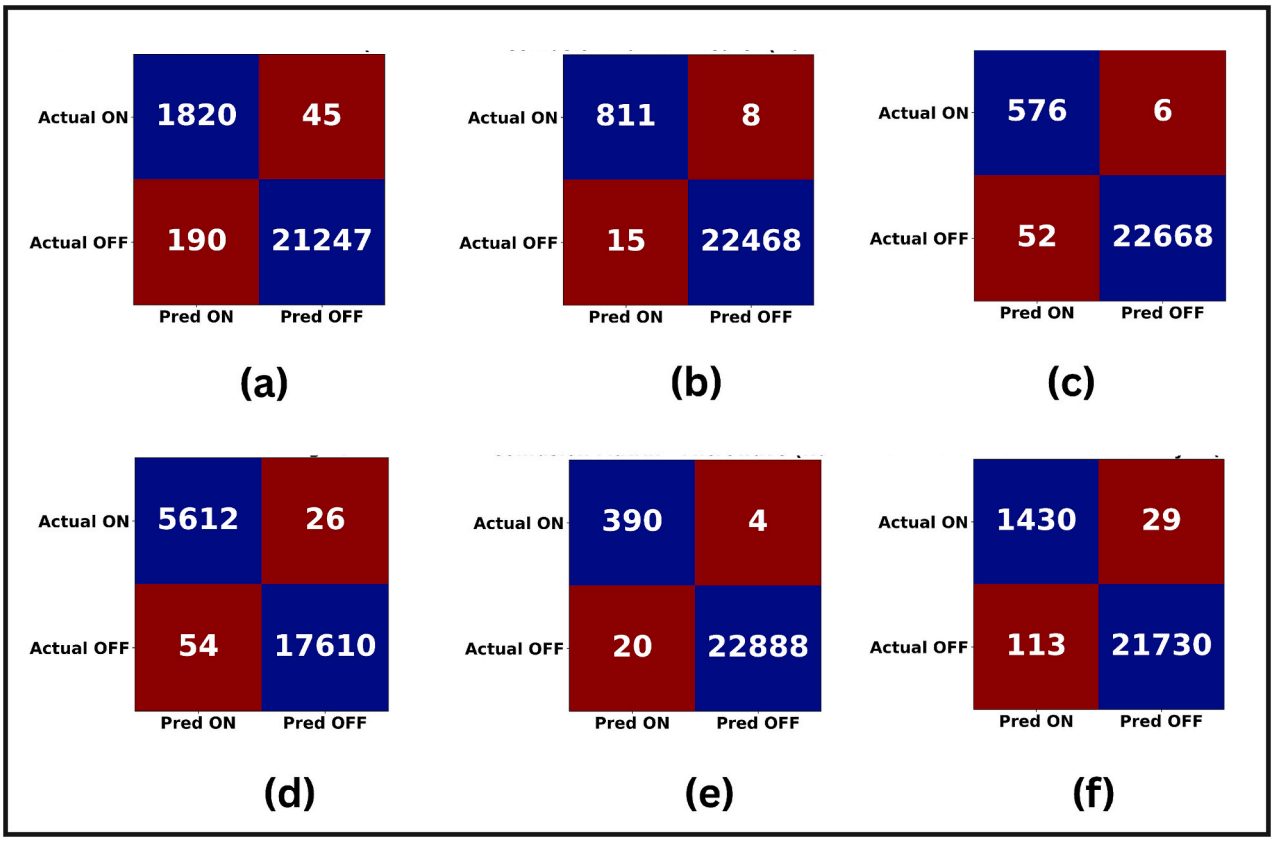



**Fig. 7.** Numerical summary of true positives, false positives, true negatives, and false negatives across all appliances a) Dishwasher (b) Electric Heater (c) Electric Stove (d) Fridge (e) Microwave (f) Washer Dryer.

pattern of the appliance, which typically increases classification difficulty, the numerical results in Fig. 7(a) confirm that 1820 ON activations were correctly detected, with only 45 missed events and 190 false positives. This underlines the model's ability to handle appliances with bursty and less frequent cycles.

The Electric Heater exhibited one of the highest levels of predictive accuracy, with 99.0% recall and 99.9% specificity, as indicated in Fig. 6 (b). The numerical distribution in Fig. 7(b) shows that 811 ON events were correctly identified with only eight missed detections and 15 false activations. This reflects the comparatively stable and distinct energy signature of the heater, which allowed the Bagged CNN to distinguish between ON and OFF states with remarkable confidence. For the Electric Stove, the Bagged CNN achieved 99.0% recall and 99.8% specificity, as presented in Fig. 6(c). The counts in Fig. 7(c) reveal that 576 ON events were correctly classified with six false negatives, while 22,668 OFF states were correctly detected with 52 false positives. These results indicate that the Bagged CNN successfully captured the transient and bursty patterns of Stove usage while maintaining low misclassification rates.

The Fridge demonstrated near-perfect classification. As shown in Fig. 6(d), the model achieved 99.5% recall for ON events and 99.7% specificity for OFF states. The raw counts in Fig. 7(d) confirm this, with 5612 correctly detected ON activations, only 26 missed events, and 54 false positives out of thousands of samples. These findings highlight the robustness of the Bagged CNN in modeling appliances with persistent, repetitive cycling behavior. The Microwave is characterized by short and bursty activations. Nevertheless, Fig. 6(e) shows that the model achieved 99.0% recall and 99.9% specificity. In Fig. 7(e), the Bagged CNN correctly classified 390 ON events while missing only four, and it recorded 22,888 true OFF predictions with 20 false positives. These results demonstrate that the Bagged CNN effectively captures transient appliance behavior while maintaining strong precision and recall.

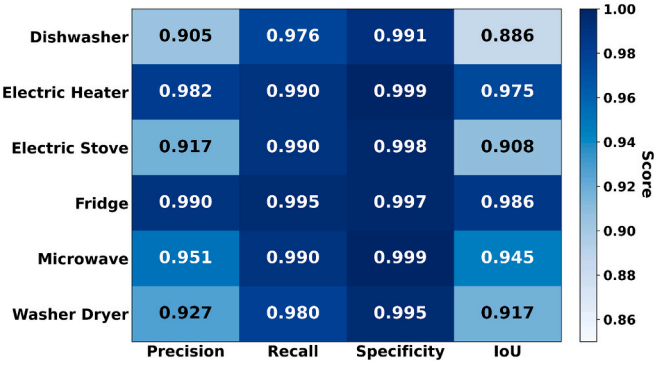



**Fig. 8.** Performance heatmap summarizing precision, recall, specificity, and intersection-over-union (IoU) across six appliances, illustrating model consistency and robustness.

Finally, the Washer Dryer showed slightly lower performance compared to the other appliances. As seen in Fig. 6(f), the model achieved 98.0% recall and 99.5% specificity. The numerical analysis in Fig. 7(f) indicates 1430 true ON detections, 29 missed events, and 113 false positives. The higher misclassification rate reflects the complexity of multi-stage operating cycles, which involve irregular transitions between washing, rinsing, and drying. Despite this, the Bagged CNN maintained strong overall performance and significantly reduced variance in predictions. In comparative terms, the Bagged CNN consistently delivered recall rates above 97% and specificity rates above 99% across all six appliances. The Fridge and Electric Heater achieved the most accurate classifications, benefiting from their regular or distinctive energy patterns, while the Washer Dryer presented the greatest challenge

due to its operational complexity. The Dishwasher and Microwave, despite sparse or transient usage, were also classified with high reliability, confirming the robustness of the ensemble approach.

To further consolidate the appliance-state classification evaluation, Fig. 8 presents a performance heatmap summarizing precision, recall, specificity, and Intersection over Union (IoU) across all six appliances. This visualization provides a holistic comparison, capturing not only detection reliability but also the balance between false positives and false negatives. The results confirm the consistency of the proposed method across appliances, with precision values ranging from 0.905 (Dishwasher) to 0.990 (Fridge). The Dishwasher recorded the lowest precision (0.905), reflecting a slightly higher false positive rate, which is consistent with its irregular and sparse activation cycles. In contrast, the Fridge achieved near-perfect precision (0.990), benefiting from its distinctive cycling pattern. Recall values were uniformly high, ranging between 0.976 and 0.995, demonstrating that the proposed method reliably captured ON events for all appliances, including those with transient behavior such as the Microwave and Electric Stove.

Specificity scores were uniformly strong across all appliances, all exceeding 0.991, indicating that OFF events were correctly classified with minimal false activations. This robustness is particularly important in NILM tasks, where OFF states typically dominate the dataset and class imbalance can bias model performance. Finally, the IoU metric provides an integrative measure of overlap between predicted and actual activations. The Dishwasher recorded the lowest IoU (0.886), consistent with its sparse ON cycles, while the Fridge achieved the highest IoU (0.986), reflecting the proposed model's strength in capturing continuous and repetitive loads. The Microwave and Washer Dryer achieved intermediate IoU scores of 0.945 and 0.917, respectively, with the latter again showing slightly reduced performance due to its multi-stage and irregular load profile. Taken together, the heatmap highlights that while all appliances benefited from ensemble averaging, the degree of improvement was most pronounced for complex or irregular devices such as the Dishwasher and Washer Dryer. In contrast, appliances with stable or cyclical patterns, such as the Electric Heater and Fridge, achieved near-optimal performance across all metrics. This reinforces the conclusion that the Bagged CNN is not only robust across heterogeneous load types but also capable of generalizing effectively to unseen household conditions.

### 3.5. Ablation and sensitivity analyses

To further verify the feasibility and robustness of the proposed CB-S2P NILM framework, additional analyses were conducted. These analyses focused on three aspects: computational tractability, component-wise contribution through ablation, and sensitivity to key temporal stabilization and ensemble parameters. Together, these experiments provide a more complete understanding of the practical cost, internal design effectiveness, and parameter robustness of the proposed method.

Computational tractability was first examined by comparing the runtime characteristics of the single Seq2Point CNN and the bagged CB-S2P variant. The comparison considered mean training time per epoch, total training time, test inference time, and parameter count, as summarized in Table 3. The single Seq2Point CNN required 3.295 s per training epoch and 59.31 s in total, with a test inference time of 4.48 s and 296,838 trainable parameters. When bagging was introduced with three independent CNN replicas, the mean training time per epoch for an individual replica remained comparable at 3.076 s. However, the total training time increased to 128.42 s because three separate models had to be optimized and retained. The bagged configuration also increased the total parameter count to 890,514 and produced a test inference time of 5.25 s. These results confirm that ensemble bagging introduces additional training cost, storage demand, and moderate inference overhead relative to a single CNN. However, the added cost arises from replication of a compact one-dimensional convolutional backbone rather than from the use of intrinsically heavier recurrent or attention-based architectures. Therefore, the proposed CB-S2P framework should be interpreted as a robustness-oriented ensemble design that trades moderate computational overhead for improved prediction stability and stronger classification performance.

**Table 3**
Runtime comparison between the single Seq2Point CNN and the bagged CB-S2P framework.

| Model | Mean training time per epoch per replica (s) | Total training time (s) | Test inference time (s) | Parameters |
|---|---|---|---|---|
| Single Seq2Point CNN | 3.295 | 59.31 | 4.48 | 296,838 |
| Bagged CB-S2P CNN (N = 3) | 3.076 | 128.42 | 5.25 | 890,514 |

Note: For the bagged CB-S2P model, the reported epoch time refers to the mean training time per epoch for one replica, whereas the total training time reflects the cost of optimizing all ensemble members.

In the ablation study, starting from a baseline Seq2Point CNN, the effects of probability calibration with device-specific threshold tuning, temporal smoothing with dwell-time constraints, and bagged ensemble prediction were incorporated progressively. The results are reported in Table 4. The baseline CNN achieved an accuracy of 0.982371 and F1-score of 0.973082. After introducing temperature scaling and device-specific threshold optimization, the average accuracy increased to 0.989324 and the average F1-score improved to 0.980336, indicating that calibration enhanced the reliability of appliance-state decisions. The addition of temporal smoothing and dwell-time constraints further improved performance, increasing the average accuracy to 0.999345 and the average F1-score to 0.993397. This indicates that temporal stabilization was effective in suppressing implausible short fluctuations, preserving legitimate appliance activations, and improving overall sequence consistency. Finally, the full CB-S2P framework, which additionally incorporated bagged ensemble prediction, achieved the highest average F1-score of 0.999428 while maintaining a high average accuracy of 0.999606. These findings confirm that the proposed framework benefits from the complementary interaction of calibration, temporal stabilization, and variance reduction, and that the full CB-S2P configuration provides the strongest overall balance of predictive performance.

For sensitivity analysis, the effects of the median filter width, minimum dwell duration, and ensemble size were examined. The results are summarized in Table 5 and Table 6. For temporal post-processing, the best performance was obtained with a median filter width of 3 and a dwell duration of 3, which achieved an average accuracy of 0.999345 and an average F1-score of 0.993397. When the dwell duration was increased to 4, the average F1-score decreased noticeably, indicating that overly strict temporal persistence constraints may suppress genuine short appliance activations. Similarly, a median filter width of 5 generally produced lower F1-scores than a width of 3, suggesting that excessive smoothing may remove useful event boundaries. These results indicate that moderate temporal stabilization offers the best balance

**Table 4**
Ablation study of the proposed CB-S2P framework.

| Variant | Description | Avg Accuracy | Avg F1 |
|---|---|---|---|
| A1 | Baseline CNN without calibration, smoothing, or bagging | 0.982371 | 0.973082 |
| A2 | A1 + temperature scaling and device-specific threshold tuning | 0.989324 | 0.980336 |
| A3 | A2 + temporal smoothing and dwell constraints | 0.999345 | 0.993397 |
| A4 | A3 + bagged ensemble prediction (full CB-S2P) | 0.999606 | 0.999428 |

**Table 5**
Sensitivity analysis for temporal smoothing and dwell-time settings.

| Median filter width | Dwell time | Avg Accuracy | Avg F1 |
|---|---|---|---|
| 3 | 2 | 0.989324 | 0.990327 |
| 3 | 3 | 0.999345 | 0.993397 |
| 3 | 4 | 0.997113 | 0.980413 |
| 5 | 2 | 0.989255 | 0.987469 |
| 5 | 3 | 0.998987 | 0.990180 |
| 5 | 4 | 0.996754 | 0.978662 |

**Table 6**
Sensitivity analysis for ensemble size.

| Ensemble size | Avg Accuracy | Avg F1 |
|---|---|---|
| 1 | 0.989473 | 0.985520 |
| 3 | 0.999606 | 0.999428 |
| 5 | 0.997817 | 0.989864 |

**Table 7**
Comparative evaluation of the proposed bagged CNN against existing NILM approaches across six appliances.

| **Dishwasher** | | | **Electric Heater** | | |
|---|---|---|---|---|---|
| **Methods** | **Accuracy** | **F1 Score** | **Methods** | **Accuracy** | **F1 Score** |
| Hierarchical modular CNN [56] | 0.94 | 0.94 | GRU-MF [54] | 0.98 | 0.94 |
| Diffusion Transfer Learning [50] | 0.98 | 0.40 | CNN-GRU [57] | 0.99 | 0.99 |
| Proposed method | 0.99 | 0.98 | Proposed method | 0.99 | 0.99 |
| **Electric Stove** | | | **Fridge** | | |
| CNN-LSTM-RF [55] | 0.95 | 0.94 | Hierarchical modular CNN [56] | 0.97 | 0.97 |
| LSTM [58] | 0.69 | 0.69 | Diffusion Transfer Learning [50] | 0.90 | 0.881 |
| Proposed method | 0.98 | 0.99 | Proposed method | 0.99 | 0.99 |
| **Microwave** | | | **Washer Dryer** | | |
| Hierarchical modular CNN [56] | 0.98 | 0.98 | CycleGAN [51] | 0.98 | 0.766 |
| SE-LSTM [48] | – | 0.87 | ResNet18 [53] | 0.97 | 0.88 |
| Proposed method | 0.99 | 0.99 | Proposed method | 0.99 | 0.98 |

between suppressing prediction chatter and preserving valid state transitions.

The ensemble-size analysis further showed that increasing the number of replicas from 1 to 3 produced a clear performance gain, with average accuracy improving from 0.989473 to 0.999606 and average F1-score increasing from 0.985520 to 0.999428. However, further increasing the ensemble size from 3 to 5 did not provide additional improvement and instead slightly reduced the average F1-score to 0.989864. This suggests that a three-replica ensemble is sufficient to achieve most of the variance-reduction benefit while avoiding unnecessary computational overhead. Overall, the sensitivity analysis confirms that the proposed CB-S2P framework performs most effectively under moderate smoothing and ensemble settings, which supports the feasibility and stability of the design choices adopted in this study.

### 3.6. Comparative analysis with existing studies

The comparative evaluation across six appliances demonstrates the consistent superiority of the proposed Bagged CNN over a range of existing NILM approaches, as summarized in Table 7. For the Dishwasher, earlier works such as the Hierarchical Modular CNN by Herath et al. [56] reported balanced but modest results with accuracy and F1 of 0.94, while Cheng et al. [50] increased accuracy to 0.98 but suffered a severe collapse in F1 to 0.40. This imbalance reflects an inability to capture sparse ON events despite reasonable overall accuracy. In contrast, proposed method reached 0.99 accuracy and 0.98 F1, showing that ensemble averaging, probability calibration, and dwell-time smoothing directly addressed the limitations of these earlier models by stabilizing predictions on irregular activations. For the Electric Heater, Putrada et al. [54] achieved 0.98 accuracy and 0.94 F1 using a GRU-MF model, while Li et al. [57] advanced performance with a CNN–GRU hybrid that achieved 0.99 for both metrics. The proposed Bagged CNN matched these state-of-the-art results, also attaining 0.99 accuracy and 0.99 F1. This comparison is notable because the proposed method achieved equivalent predictive performance using a structurally simpler Seq2Point CNN-based ensemble, without relying on recurrent units or time–frequency image fusion. Although Li et al. [57] reported that 1D-CNN exhibited faster convergence during training than both 2D-CNN and GRU.

The Electric Stove further illustrates the advantage of the proposed approach. Fang et al. [58] reported weak results with an LSTM model (0.69 accuracy and 0.69 F1), which struggled to handle short bursty activations. Zhou et al. [55] improved these outcomes using a CNN–LSTM–RF hybrid, reaching 0.95 accuracy and 0.94 F1, though at the cost of increased architectural complexity. The Bagged CNN surpassed both, with 0.98 accuracy and 0.99 F1, confirming that ensemble averaging among CNNs can outperform hybrid designs by reducing variance and limiting false activations, while maintaining high recall for transient events. In the case of the Fridge, Herath et al. [56] achieved strong results with 0.97 accuracy and F1, reflecting the advantage of modular CNNs in modeling structured cycling signals. However, Cheng et al. [50] performed less effectively, with only 0.90 accuracy and 0.881 F1, showing the difficulty of applying transfer learning to appliances with frequent micro-cycles. The Bagged CNN achieved 0.99 accuracy and 0.99 F1, outperforming both prior approaches. This performance is tied to its preprocessing pipeline and temporal smoothing, which effectively suppressed chatter and preserved genuine transitions, allowing it to reliably model repetitive cycling behavior.

The Microwave illustrates how transient loads benefit from ensemble averaging. Herath et al. [56] obtained competitive results with 0.98 accuracy and F1 using a modular CNN, while Hwang and Kang [48] achieved only 0.87 F1 with an SE-LSTM, which was ill-suited to short-duration events. The proposed Bagged CNN achieved the highest scores (0.99 accuracy and 0.99 F1), showing its ability to generalize to irregular and bursty activations that typically lead to misclassifications. This superior performance is directly linked to the combination of $\Delta x$-based feature construction, calibration, and bagging, which jointly enhance sensitivity to short events while maintaining low false positive rates. Finally, the Washer Dryer presented the greatest challenge due to its multi-stage cycles. Yang et al. [51] applied CycleGAN, which achieved 0.98 accuracy but a low F1 of 0.766, while Li et al. [53] reported 0.97 accuracy and 0.88 F1 with ResNet18. Both models struggled to balance precision and recall under irregular wash–rinse–dry transitions. The Bagged CNN significantly improved performance, reaching 0.99 accuracy and 0.98 F1, confirming its ability to stabilize predictions across irregular sequences. Bagging across multiple CNN replicas minimized variance and reduced misclassifications, allowing the model to outperform both generative and residual-based baselines.

The superior performance of the proposed framework can be attributed to the methodological design choices that directly address long-standing challenges in minute-level state disaggregation. First, the

adoption of θ-validated threshold calibration ensured that appliance labels were derived through a systematic grid-search over validation data rather than by heuristic power cut-offs. This eliminated inconsistencies in the training supervision, leading to more reliable learning of device activation boundaries. Second, the weighted binary cross-entropy formulation with per-appliance balancing factors effectively mitigated the class imbalance inherent in NILM datasets, allowing the model to remain sensitive to rare ON events without sacrificing OFF-state accuracy. Third, the bagging of multiple independent CNN replicas substantially reduced the variance arising from random initialization and small dataset size. The ensemble averaging of calibrated probabilities produced smoother and more stable outputs, suppressing the stochastic fluctuations often observed in single-network predictions. Fourth, the temperature-scaling calibration step refined the probabilistic confidence of each appliance output, aligning model scores with true empirical frequencies and improving threshold selection. Finally, the inclusion of temporal smoothing and dwell-time constraints imposed realistic continuity on appliance state sequences, removing spurious chatter and ensuring temporal coherence.

However, the proposed framework remains dependent on manually selected hyperparameters, including the median filter width and dwell-time duration, which may require re-optimization when applied to datasets with different sampling characteristics or appliance dynamics. In addition, the framework assumes that appliance-specific calibration parameters and decision thresholds estimated from validation data remain transferable to the testing phase. While the strong cross-house results obtained on REDD suggest that this assumption is reasonable within the statistical range of that dataset, it may become less reliable under more pronounced domain shifts arising from differences in appliance composition, household usage behavior, metering noise, sampling rate, or regional load characteristics. In such cases, dataset-specific recalibration, adaptive threshold refinement, or transfer-learning-based adaptation may be necessary. Accordingly, although the proposed framework exhibits strong generalization within REDD, broader validation on datasets such as UK-DALE, REFIT, and Pecan Street is required to establish the robustness of its calibration and thresholding strategy under more heterogeneous practical conditions.

## 4. Conclusion

This study developed and evaluated a calibrated bagged Seq2Point one-dimensional convolutional neural network framework, denoted as CB-S2P NILM, for robust appliance-state disaggregation under low-frequency smart-meter conditions. The proposed framework was designed to address several practical weaknesses that remain common in NILM, namely heuristic label generation, severe ON/OFF imbalance, poorly calibrated output probabilities, temporally fragmented predictions, and variance across repeated model runs. To overcome these limitations, the study integrated appliance-specific threshold validation, class-balanced learning, temperature scaling, temporal smoothing with dwell-time constraints, and bagged ensemble prediction into a unified state-classification framework.

The experimental results on the REDD cross-house setting showed that the proposed framework achieved strong and consistent performance across six appliances. During validation, the Bagged CNN improved appliance-level accuracy from 0.93 to 0.96 for the baseline CNN to 0.98–0.99, while F1-score increased from 0.91 to 0.96 to 0.98–0.99 across all evaluated appliances. In the ablation analysis, the baseline configuration achieved an average accuracy of 0.982371 and an average F1-score of 0.973082, whereas the full CB-S2P framework increased these values to 0.999606 and 0.999428, respectively. Sensitivity analysis further showed that an ensemble size of three provided the best overall balance, outperforming both a single model and a larger five-model ensemble. Appliance-wise testing results also remained consistently strong, with recall values above 97% and specificity above 99% across all six appliances, confirming robust generalization to an unseen house.

In addition to predictive performance, the study also examined the computational implications of the proposed framework. The single Seq2Point CNN required a mean training time of 3.295 s per epoch, a total training time of 59.31 s, and a test inference time of 4.48 s, with 296,838 parameters. By comparison, the bagged CB-S2P model with three ensemble members required a mean training time of 3.076 s per epoch per replica, a total training time of 128.42 s, and a test inference time of 5.25 s, with 890,514 parameters. These results show that the improved classification reliability was obtained with a moderate increase in training cost and only a small increase in inference time, which supports the practical feasibility of the framework for research-oriented and low-frequency smart-meter applications.

These findings demonstrate that the contribution of this work is not only the use of a compact Seq2Point CNN, but the systematic combination of calibrated label construction, probability correction, temporal post-processing, and ensemble averaging within a single NILM pipeline. The results show that reliable state disaggregation does not necessarily require heavier recurrent, transformer, or time–frequency fusion architectures. Instead, a compact convolutional design can deliver highly competitive performance when its prediction process is carefully calibrated and temporally stabilized. In this respect, the study contributes a practical and reproducible framework for improving reliability, interpretability, and robustness in low-frequency NILM.

Despite these promising outcomes, several directions remain open for future research. The current framework was evaluated only on the REDD dataset and on six selected appliances, so broader validation on datasets such as UK-DALE, REFIT, and Pecan Street is still required to examine transferability under different household characteristics, appliance compositions, and sampling conditions. Future work should also investigate adaptive calibration strategies that reduce dependence on fixed validation-derived thresholds and smoothing parameters when the data distribution changes. In addition, extending the framework toward cross-domain generalization, lightweight deployment, and joint state-and-energy estimation would further strengthen its practical value for real-world NILM applications.

## Declaration of Generative AI and AI-assisted technologies in the writing process

Throughout the process of crafting this work, the authors have utilized ChatGPT to enhance clarity and language. Following the utilization of this tool/service, the authors have carefully reviewed and edited the content as required, assuming complete responsibility for the publication's substance.

## CRediT authorship contribution statement

**Md Mahadi Hasan Imran:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Muhamad Zalani Daud:** Writing – review & editing, Visualization, Supervision, Methodology. **Muhammad Mukhtar Qureshi:** Writing – review & editing, Software, Formal analysis. **Shahrizan Jamaludin:** Writing – review & editing, Supervision. **Atta-ur-Rahman:** Writing – review & editing, Supervision, Formal analysis. **Md Meherullah:** Visualization, Validation, Software, Formal analysis. **Abdullah Alqahtani:** Writing – review & editing, Validation, Supervision, Methodology.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

The authors acknowledge the financial support from the Ministry of Higher Education Malaysia (MOHE) through the Fundamental Research Grant Scheme (FRGS/1/2024/TK07/UMT/02/1).

## Appendix 1. Principal hyperparameter settings of the proposed CB-S2P NILM framework

| Component | Hyperparameter | Value |
|---|---|---|
| Input representation | Window length (W) | 128 |
| Input representation | Number of input channels (C) | 4 |
| Input representation | Feature channels | Raw mains, first-order difference, rolling mean, rolling standard deviation |
| Input representation | Rolling statistics window (ω) | [5, 7] |
| Windowing | Stride | 1 |
| CNN architecture | Number of Conv1D blocks | 5 |
| CNN architecture | Channel widths | [64, 64, 128, 128, 256] |
| CNN architecture | Kernel sizes | [5, 5, 7, 7, 3] |
| CNN architecture | Dilations | [1, 2, 2, 4, 4] |
| CNN architecture | Dropout rates | [0.10, 0.10, 0.15, 0.15, 0.20] |
| CNN architecture | Activation function | ReLU |
| CNN architecture | Pooling | Global average pooling |
| Training | Optimizer | Adam |
| Training | Initial learning rate | 0.001 |
| Training | Minimum learning rate | 0.0001 |
| Training | Weight decay (λ) | 0.0001 |
| Training | Batch size | 256 |
| Training | Maximum epochs | 100 |
| Training | Early stopping patience | 10 |
| Training | ReduceLROnPlateau patience | 5 |
| Training | ReduceLROnPlateau factor | 0.5 |
| Loss function | Primary objective | Weighted binary cross-entropy |
| Loss function | Alternative objective in ablation | Focal loss |
| Thresholding | Decision threshold search range | 0.01 to 0.99 |
| Temporal smoothing | Median filter candidates | [3, 5] |
| Duration constraints | Minimum ON duration candidates | [2, 3, 4] |
| Duration constraints | Minimum OFF duration candidates | [2, 3, 4] |
| Ensemble learning | Ensemble type | Homogeneous bagging |
| Ensemble learning | Main experiment ensemble size | 3 |

## Appendix A2. Comparative description of existing NILM approaches used for benchmarking, including datasets, appliances tested, methodological highlights, and achieved performance metrics

**Method and details**

**Name of the Approach**: Hierarchical modular CNN-based NILM anomaly detection [47].
**Dataset**: REDD dataset (Houses 1–3); appliances: refrigerator, microwave, dishwasher.
**Appliance(s) used for comparison**: Dishwasher, Fridge, Microwave.
**Description**: The methodology adopts a plug-and-play hierarchical architecture designed to simplify the integration of new appliances and fault categories while maintaining high computational efficiency. The overall framework is composed of three sequential CNN modules: the Appliance Identification CNN (AICNN) identifies active appliances from aggregate signals using a parallel-path convolutional structure with multiple kernel sizes; the Fault Identification CNN (FICNN) distinguishes normal from faulty operation; and, upon detecting a fault, the Fault Type Identification CNN (FTICNN) classifies the specific anomaly type, such as elongated duty cycles or over-amplitude consumption. The proposed system achieved consistently strong performance across all appliances, with Stage 1 (appliance detection and fault identification) yielding high accuracy, precision, recall, and F1-scores for refrigerator (0.97), microwave (0.98), and dishwasher (0.94), while Stage 2 (anomaly classification) successfully identified nearly all inserted anomalies, except for a few elongated dishwasher and refrigerator cycles where signatures were visually similar. Comparative analysis further confirmed its superiority over existing NILM anomaly detection frameworks, outperforming methods by Hosseini et al. [59], Quek et al. [60], and Rashid et al. [61], as the hierarchical CNN consistently maintained balanced detection rates and demonstrated the capability to identify multiple fault types simultaneously, an aspect where prior models often underperformed.
**Name of the Approach**: SE-LSTM (State Encoding LSTM with Feedback Structure) [48].
**Dataset**: REDD dataset (low-frequency, 1-min intervals) and Living-lab dataset (real-world validation with TV, rice cooker, washer Dryer, microwave, lighting).
**Appliance(s) used for comparison**: Microwave.
**Description**: The methodology addresses two key challenges in low-resolution NILM: the loss of appliance usage patterns and severe class imbalance. To overcome these issues, the SE-LSTM introduces a feedback structure in which predictions from previous time steps are reintroduced into the network through an encoding block that initializes and enriches the hidden and cell states of the LSTM, thereby preserving temporal context and enhancing its ability to learn minority appliance behaviors. A select layer filters weak activations and substitutes them with encoded states to prevent the neglect of important usage histories. The training consists of two stages, standard optimization via backpropagation through time and threshold adjustment for the select layer using natural evolution strategies (NES), which efficiently replaces exhaustive search. Unlike oversampling methods such as SMOTE, SE-LSTM learns directly from imbalanced data, maintaining robustness without generating synthetic samples. Experimental results show that the model consistently outperforms baseline methods, achieving an F1-score of 0.86 for the oven in REDD House 1 compared to 0.0

(*continued on next page*)

(*continued*)

and 0.25 for LSTM + SMOTE and LSTM + Encoding Network, respectively, while maintaining F1 values between 0.6 and 0.9 across all appliances. Validation on the Living-lab dataset further confirmed its practical efficiency, reaching 0.8–0.9 F1 with a runtime of only one second per 128 mini-batch on GPU. Overall, SE-LSTM outperformed competing techniques such as SVMs, CNN + GAF, GRU, and stateful LSTMs by effectively capturing device usage sequences even under data imbalance, making it especially effective for low-frequency NILM applications.

**Name of the Approach**: TransUNet-NILM (ResNet + Transformer + U-Net hybrid) [49].

**Dataset**: REDD dataset (4 appliances: refrigerator, washing machine, microwave oven, Dryer); U.K.-DALE dataset (5 appliances: kettle, refrigerator, washing machine, microwave oven, Dryer).

**Appliance(s) used for comparison**: Fridge, Microwave.

**Description**: The methodology addresses the limitations of earlier CNN, RNN, and Transformer-based NILM models by integrating local and global feature extraction for low-frequency and noisy aggregate power data. NILM is formulated as a sequence learning problem where the encoder employs ResNet blocks with residual connections to capture local appliance features, while a Transformer block with multi-head self-attention models long-term dependencies across time. These features are combined through a U-Net style encoder-decoder with skip connections to reconstruct appliance-level power signals, and the decoder uses deconvolutional and linear layers optimized by mean squared error loss with early stopping. Experimental results show that TransUNet-NILM achieved 14.9 percent and 18.9 percent higher F1-scores on the REDD and UK-DALE datasets respectively, demonstrating superior noise resilience, lower MAE and SAE, and greater computational efficiency compared to Transformer, CNN-BiGRU, and seq2point models.

**Name of the Approach**: Diffusion Model with Transfer Learning (Diffusion-TL NILM) [50].

**Dataset**: REDD dataset (source domain: multiple appliances from Houses 1–3) and REFIT dataset (20 UK households, 8-second sampling; target domain).

**Appliance(s) used for comparison**: Dishwasher, Fridge.

**Description**: The methodology addresses the limitations of conventional NILM models that depend on large, labeled datasets and perform poorly when transferred across households with different appliance usage distributions. To overcome these issues, the Diffusion-TL NILM framework integrates diffusion probabilistic modeling with few-shot transfer learning, where a forward process progressively adds Gaussian noise to appliance signals and a reverse process, modeled by a neural network with residual layers, bidirectional dilated convolutions, skip connections, and gating units, reconstructs clean load signals by predicting and refining noise distributions at each step. The model is first trained on the REDD dataset to learn general load patterns and then fine-tuned on the REFIT dataset, updating only the initial convolutional and fully connected layers while keeping the deeper residual layers fixed to retain shared representations. This selective adaptation enhances transfer efficiency and eliminates the need for full retraining. Experimental results confirm that Diffusion-TL NILM surpasses the CNN-based transfer learning baseline of D'Incecco et al. [62], with F1-score improvements from 0.55 to 0.88 for refrigerator, 0.36 to 0.67 for washing machine, and 0.16 to 0.42 for microwave, indicating higher accuracy, precision, and robustness across diverse building domains.

**Name of the Approach**: Adversarial Anomaly Detection & Calibration (Cycle-Consistent GAN) [51].**Dataset**: REDD dataset (5 houses, 119 days, appliances: Fridge, washer Dryer, microwave), UK-DALE dataset (5 homes, appliances: kettle, Fridge, washer Dryer, microwave, dishwasher), CLP dataset (Hong Kong; air conditioners, washing machine).

**Appliance(s) used for comparison**: Washer Dryer.

**Description**: The methodology addresses a gap in existing NILM research, where most efforts focused on improving neural network structures but overlooked the influence of abnormal data (both point-type and pattern-type anomalies) that frequently occur in smart meter recordings. The approach is built on an enhanced cycle-consistent generative adversarial network (CycleGAN) that encodes aggregated load measurements into a low-dimensional latent space and reconstructs them back. Abnormal data, which deviate from typical load patterns, produce high reconstruction errors, making them identifiable. The detection process is performed using three discriminators that regularize the cycle-consistency approximation between real data and generated signals, ensuring robustness against anomalies. After anomaly detection, the method applies a calibration step in the latent space. Using DBSCAN clustering and nearest neighbor approximation, the framework replaces anomalous points with calibrated values based on nearby normal data distributions. This ensures that both point-type anomalies (isolated spikes) and pattern-type anomalies (continuous corrupted sequences) are corrected. Results showed consistent improvements across different NILM backbones, including Transformer, CNN, LSTM, BiLSTM, and TCN.

**Name of the Approach**: Seq2-[3]point CNN + Transfer Learning; 2D-CNNs for Appliance Identification [52].

**Dataset**: REDD dataset (low-frequency, 1 Hz, 119 days, 6 houses) for home-NILM; REFIT dataset (20 homes, 8-s sampling) for site-NILM; custom 2D dataset of wavelet/STFT/fused spectrograms for appliance identification.

**Appliance**(s) used for comparison: Washer Dryer.

**Description**: The authors propose a transfer learning framework for NILM and appliance identification, extending the widely used seq2point CNN model into a seq2-[3]point CNN and leveraging 2D-CNNs for appliance signature classification. For home-NILM, the seq2-[3]point CNN model is designed to take a window of aggregate mains power as input and output three points of the corresponding appliance load (start, midpoint, and end). This extension captures more temporal context than seq2point while maintaining computational efficiency. The network consists of five convolutional layers followed by two fully connected layers. After training on one appliance, the convolutional layers are frozen, and transfer learning is applied to other appliances, reducing training cost and enabling reuse of learned features across devices. Results showed strong performance, with the highest accuracy reaching 94.6% for dishwasher, 94.4% for microwave, 86.6% for refrigerator, and 89.9% for washer Dryer, clearly outperforming earlier seq2point implementations.

**Name of the Approach**: CNN with Mixed Residual Unit [53].

**Dataset**: Custom dataset collected from 22 electrical appliances (voltage & current signals at 4 kHz; each appliance ON/OFF at least 100 times; 1000 VI trajectory images per appliance, 64 × 64 resolution).

**Appliance(s) used for comparison**: Electric Heater, Washer Dryer.

**Description**: The methodology is motivated by the limitations of traditional feature engineering and CNN-based NILM, where deeper networks often suffer from gradient vanishing or exploding, leading to poor classification accuracy for multi-state appliances. The approach begins with data pre-treatment, where voltage and current signals from 22 different electrical appliances were collected at a sampling rate of 4 kHz. Each appliance was switched on and off at least 100 times, generating transient data that was converted into VI trajectory images. Each trajectory image had a size of 64 × 64 pixels, with 1000 images generated per appliance (800 for training and 200 for testing). The CNN architecture integrates residual units into a traditional CNN framework to prevent gradient dispersion in deep layers. The model structure includes an initial convolutional layer (3 × 3 kernel), pooling, a residual unit, additional convolution and pooling layers, followed by two fully connected layers (1024 hidden units and an output layer of 22 units). The residual

(*continued on next page*)

(*continued*)

mapping (H(x) = F(x) + x) allows the network to better approximate identity mappings and improves training stability. The results demonstrated high classification performance, achieving an overall accuracy of 97.9% across the 22 appliances. When compared to earlier NILM approaches, such as VI trajectory-based CNN models (De Baets et al., 2018), which reported accuracies around 75–78% on PLAID and WHITED datasets, the proposed mixed residual CNN significantly improved performance.

**Name of the Approach**: GRU-MF (Gated Recurrent Unit with Mode Filtering) [54].

**Dataset**: Smart Home Dataset with Weather Information (Kaggle). Four appliances used: dishwasher, heater, refrigerator, and lighting (10,000 samples each).

**Appliance(s) used for comparison**: Electric Heater.

**Description**: The methodology builds on the strengths of gated recurrent units (GRU) for sequence modeling and incorporates a mode filtering (MF) step to smooth prediction results. The preprocessing pipeline was divided into two stages. The first stage applied autocorrelation and time-series transformation to prepare the sequential data. The second stage involved normalization, standardization, label encoding, and one-hot encoding to structure the dataset for deep learning. Time-step selection was determined through ADF, ACF, and PACF tests, with lag = 20 found optimal. Results showed that the GRU outperformed all benchmarks, achieving accuracy = 0.96, higher than CNN and DNN (0.94), LSTM (0.93), and RNN (0.91). Precision and recall analysis further highlighted GRU's superiority: dishwasher recall reached 0.99, while heater recall was 0.89 due to partial misclassification with refrigerator signals. A second test set of 3000 items yielded an overall accuracy of 0.86. In comparison to earlier NILM works, such as Zhang et al. [63] with CNN (accuracy = 0.97), de Paiva Penha et al. [64] with DNN (accuracy = 0.93), and Kim et al. [65] with LSTM (accuracy = 0.95)—the GRU-MF approach delivered superior results.

**Name of the Approach**: Time-Frequency Feature Fusion with Two-Stream CNN-GRU [53].

**Dataset**: PLAID dataset (11 appliances, 30 kHz sampling, 56 homes; ~1793 instances), IDOUC dataset (103 appliance types, 4 kHz, 100 samples each, steady-state + transient).

**Appliance(s) used for comparison**: Electric Heater.

**Description**: The methodology is specifically designed to improve the identification of multistate and low-power appliances, which are often difficult to distinguish using single-domain features. The process begins with data preprocessing, where steady-state current signals are extracted from raw NILM data. These signals are converted into two-dimensional images using multiple encoding algorithms: Gramian Angular Summation Field (GASF), Gramian Angular Difference Field (GADF), Markov Transition Field (MTF), and a newly introduced current spectrum (CS) diagram. The fusion of these representations produces a four-channel image that simultaneously retains static, dynamic, and spectral features of the appliance signals. This ensures that both time-domain and frequency-domain information are preserved. To classify appliances, the authors design a two-stream convolutional neural network (CNN) that integrates a GRU stream (for sequential sensitivity) and a 2D-CNN stream (for image-based feature extraction). The outputs from both streams are fused using a feature transformation and regression weighting mechanism, which separates independent features from shared ones before final classification with a multilayer perceptron. This approach not only improves recognition accuracy but also accelerates training convergence compared to standalone CNNs. Results demonstrated that the proposed approach significantly outperformed existing NILM models, achieved an accuracy of 99.49% and F1-macro of 0.9939, compared to 80.7% accuracy and 0.7871 F1 for the state-of-the-art baseline.

**Name of the Approach**: CNN-LSTM-RF (Convolutional Neural Network + Long Short-Term Memory + Random Forest) [55].

**Dataset**: REDD dataset (6 houses, appliances: AC, Fridge, dishwasher, washer/Dryer, microwave, Stove); Pecan Street dataset (nearly 1000 homes, circuit-level minute data for AC, Fridge, EV, washer, dishwasher, PV).

**Appliance(s) used for comparison**: Electric Stove.

**escription**: The methodology begins with a multi-feature and high-dimensional method (MFHDM), which transforms 1D low-frequency power data into 2D matrices by extracting statistical features such as mean, variance, difference, and sum over multiple time periods (1 min, 5 min, 30 min, 1 h, 1 day). The deep learning model integrates three components: CNN for feature extraction from the MFHDM matrices, LSTM for modeling temporal dependencies and long-term appliance usage patterns, and Random Forest (RF) for decoding outputs and determining appliance states. RF serves as a post-processor, setting adaptive thresholds to convert probability outputs into ON/OFF states and improving classification stability. To address class imbalance, a compound re-weighting method adjusts label weights during training based on their pre-training recognition accuracy, ensuring minority appliances are not overlooked. The model achieved 95% accuracy on REDD and 94% accuracy on Pecan Street, outperforming KNN, CO, FHMM, CNN-LSTM, and PBN baselines.

**Name of the Approach**: Appliance-Specific Lightweight LSTM Networks with Bayesian Non-Parametric State Extraction [55].**Dataset**: REDD dataset (House 1: oven, refrigerator, dishwasher, microwave, washer Dryer; House 2: refrigerator, microwave, Stove, dishwasher, kitchen outlet).

**Appliance(s) used for comparison**: Electric Stove.

**Description**: The authors propose appliance-specific lightweight LSTM networks with Bayesian non-parametric state extraction for NILM, aiming to improve accuracy while keeping the model computationally efficient. The methodology consists of three main components. First, an unsupervised Bayesian non-parametric learning approach is used for appliance signature state extraction. This leverages the Beta-process autoregressive hidden Markov model (BP-AR-HMM) to automatically determine the number and patterns of appliance operating states without requiring manual labeling. Second, the authors introduce a differential sample generation process, where the input to the model is not the raw aggregate load but the difference between pre- and post-switching events. This simplifies the learning task by focusing only on changes in appliance states rather than absolute power levels. It makes the model more robust to overlapping appliance activity and reduces the complexity of training. Third, to address the class imbalance problem, especially for rare switching events of high-capacity appliances, the SMOTE (Synthetic Minority Over-sampling Technique) algorithm is applied. By generating synthetic minority samples in the new differential feature space, the method ensures balanced training and prevents the model from ignoring rare but important events. For classification, the framework uses appliance-specific lightweight LSTM networks, each dedicated to one appliance. These networks consist of two LSTM layers (16 units each) and a fully connected Softmax output layer, keeping the architecture compact while effectively modeling temporal dependencies. Cross-validation showed strong results: on House 1, the average F-measure reached 0.676, with RMSE of 858.10 W and MAE of 75.51 W; on House 2, the method achieved an average F-measure of 0.69.

## Data availability

Data will be made available on request.

## References


[1] P.A. Schirmer, I. Mporas, Non-intrusive load monitoring: a review, IEEE Trans. Smart Grid 14 (2023), https://doi.org/10.1109/TSG.2022.3189598.

[2] Lei Y, Feng Z, Lin X, Cao D, Zhang J. NILM-LANN: A Lightweight Attention-based Neural Network in Non-Intrusive Load Monitoring. Proceedings of the 2024 27th International Conference on Computer Supported Cooperative Work in Design, CSCWD 2024, Institute of Electrical and Electronics Engineers Inc.; 2024, p. 181–6. https://doi.org/10.1109/CSCWD61410.2024.10580201.

[3] Y. Lu, S. Xu, Y. Liu, X. Jiang, Federated learning-enhanced generative models for non-intrusive load monitoring in smart homes, Sci. Rep. 15 (2025), https://doi.org/10.1038/s41598-025-11403-1.

[4] A. Pirnat, B. Bertalanič, G. Cerar, M. Mohorčič, C. Fortuna, Energy efficient deep multi-label ON/OFF classification of low frequency metered home appliances, IEEE Access 12 (2024) 51966–51981, https://doi.org/10.1109/ACCESS.2024.3382830.

[5] G.F. Angelis, C. Timplalexis, S. Krinidis, D. Ioannidis, D. Tzovaras, NILM applications: Literature review of learning approaches, recent developments and challenges, Energy Build. 261 (2022), https://doi.org/10.1016/j.enbuild.2022.111951.

[6] D. Precioso, D. Gómez-Ullate, Thresholding methods in non-intrusive load monitoring, J. Supercomput. 79 (2023), https://doi.org/10.1007/s11227-023-05149-8.

[7] O. Al-Khadher, A. Mukhtaruddin, F. Ridzuan Hashim, M.M. Azizan, H. Mamat, An implementation framework overview of non-intrusive load monitoring, J. Sustaina Dev. Energy, Water Environ. Syst. 11 (2023). https://doi.org/10.13044/j.sdewes.d11.0471.

[8] L. Massidda, M. Marrocu, S. Manca, Non-intrusive load disaggregation by convolutional neural network and multilabel classification, Appl. Sci. (Switzerland) 10 (2020), https://doi.org/10.3390/app10041454.

[9] Bansal V, Khoiwal R, Shastri H, Khandor H, Batra N. "I do not know": Quantifying Uncertainty in Neural Network Based Approaches for Non-Intrusive Load Monitoring. BuildSys 2022 - Proceedings of the 2022 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, 2022. https://doi.org/10.1145/3563357.3564063.

[10] M.M.H. Imran, S. Jamaludin, M.I. Khan, A. ur Rahman, A.F.M. Ayob, M.Z. Bin Daud, et al., Ship structure corrosion detection using advanced image processing, active contour algorithm, and parallel processing, Ocean Eng. 341 (2025), https://doi.org/10.1016/j.oceaneng.2025.122351.

[11] M.M.H. Imran, S. Jamaludin, A.F. Mohamad Ayob, A critical review of machine learning algorithms in maritime, offshore, and oil & gas corrosion research: a comprehensive analysis of ANN and RF models, Ocean Eng. 295 (2024), https://doi.org/10.1016/j.oceaneng.2024.116796.

[12] L.D.S. Nolasco, A.E. Lazzaretti, B.M. Mulinari, DeepDFML-NILM: a new CNN-based architecture for detection, feature extraction and multi-label classification in NILM signals, IEEE Sens. J. 22 (2022) 501–509, https://doi.org/10.1109/JSEN.2021.3127322.

[13] R. Ma, W. Kang, J. Sha, J. Liu, D. Zhu, NILM load identification based on CNN and V-I trajectory, in: 2024 IEEE 2nd International Conference on Power Science and Technology, ICPST 2024, Institute of Electrical and Electronics Engineers Inc, 2024, pp. 2079–2084, https://doi.org/10.1109/ICPST61417.2024.10601881.

[14] J. Chen, X. Wang, X. Zhang, W. Zhang, Temporal and spectral feature learning with two-stream convolutional neural networks for appliance recognition in NILM, IEEE Trans. Smart Grid 13 (2022) 762–772, https://doi.org/10.1109/TSG.2021.3112341.

[15] Papageorgiou PG, Christoforidis GC, Bouhouras AS. A Novel Time-Driven Twin CNN-Based NILM Framework Using Double Grayscale Images of Odd Harmonic Currents. Proceedings - 24th EEEIC International Conference on Environment and Electrical Engineering and 8th I and CPS Industrial and Commercial Power Systems Europe, EEEIC/I and CPS Europe 2024, Institute of Electrical and Electronics Engineers Inc.; 2024. https://doi.org/10.1109/EEEIC/ICPSEurope61470.2024.10751640.

[16] Y. Zhang, H. Wu, Q. Ma, Q. Yang, Y. Wang, A learnable image-based load signature construction approach in NILM for appliances identification, IEEE Trans. Smart Grid 14 (2023) 3841–3849, https://doi.org/10.1109/TSG.2023.3239598.

[17] M. Tapiador, L. De Diego-Oton, A. Hernandez, R. Nieto, Implementing a CNN in FPGA programmable logic for NILM application, in: 2023 38th Conference on Design of Circuits and Integrated Systems, DCIS 2023, 2023, https://doi.org/10.1109/DCIS58620.2023.10335989.

[18] Edmonds J, Abdallah ZS. IMG-NILM: A Deep learning NILM approach using energy heatmaps 2022. https://doi.org/10.48550/arXiv.2207.05463.

[19] E.L. de Aguiar, A.E. Lazzaretti, B.M. Mulinari, D.R. Pipa, Scattering transform for classification in non-intrusive load monitoring, Energies (Basel) 14 (2021), https://doi.org/10.3390/en14206796.

[20] R.V.A. Monteiro, J.C.R. de Santana, R.F.S. Teixeira, A.S. Bretas, R. Aguiar, C.E. P. Poma, Non-intrusive load monitoring using artificial intelligence classifiers: performance analysis of machine learning techniques, Electr. Pow. Syst. Res. 198 (2021), https://doi.org/10.1016/j.epsr.2021.107347.

[21] E.L. De Aguiar, N.L. Da Silva, A.E. Lazzaretti, D.R. Pipa, H.S. Lopes, ST-NILM: a wavelet scattering-based architecture for feature extraction and multilabel classification in NILM signals, IEEE Sens. J. 24 (2024) 10540–10550, https://doi.org/10.1109/JSEN.2024.3360188.

[22] P. Papageorgiou, D. Mylona, K. Stergiou, A.S. Bouhouras, A time-driven deep learning NILM framework based on novel current harmonic distortion images, Sustainability (Switzerland) 15 (2023), https://doi.org/10.3390/su151712957.

[23] Belguermi Y, Wira P, Hermann G. A multi-output LSTM-CNN learning scheme for power disaggregation within a NILM framework. IEEE International Conference on Industrial Informatics (INDIN), vol. 2023- July, Institute of Electrical and Electronics Engineers Inc.; 2023. https://doi.org/10.1109/INDIN51400.2023.10217936.

[24] H. Grover, L. Panwar, A. Verma, B.K. Panigrahi, T.S. Bhatti, A multi-head Convolutional Neural Network based non-intrusive load monitoring algorithm under dynamic grid voltage conditions, Sustainable Energy Grids Networks 32 (2022), https://doi.org/10.1016/j.segan.2022.100938.

[25] N. Lin, B. Zhou, G. Yang, S. Ma, Multi-head attention networks for nonintrusive load monitoring, in: ICSPCC 2020 - IEEE International Conference on Signal Processing, Communications and Computing, Proceedings, Institute of Electrical and Electronics Engineers Inc, 2020, https://doi.org/10.1109/ICSPCC50002.2020.9259533.

[26] Y. Cheng, Y. Zhong, Non-intrusive load monitoring based on a combination of transformer and CNN, in: 2024 4th International Conference on Energy, Power and Electrical Engineering, EPEE 2024, Institute of Electrical and Electronics Engineers Inc, 2024, pp. 417–421, https://doi.org/10.1109/EPEE63731.2024.10875106.

[27] D. Zhang, X. Li, J. Li, G. Li, N. Zheng, Z. Wang, Non-invasive load monitoring based on CNN-LSTM-attention, in: Proceedings of 2024 IEEE 7th International Electrical and Energy Conference, CIEEC 2024, Institute of Electrical and Electronics Engineers Inc, 2024, pp. 2536–2542, https://doi.org/10.1109/CIEEC60922.2024.10583635.

[28] J. Feng, K. Li, H. Zhang, X. Zhang, Y. Yao, Multichannel spatio-temporal feature fusion method for NILM, IEEE Trans Industr Inform 18 (2022) 8735–8744, https://doi.org/10.1109/TII.2022.3148297.

[29] H. Wu, Y. Ruan, H.Q. TimeNILM, Time-aware dual-CNN model for non-intrusive load monitoring, in: Proceedings - 2024 International Conference on Artificial Intelligence of Things and Systems, AIoTSys 2024, Institute of Electrical and Electronics Engineers Inc, 2024, https://doi.org/10.1109/AIoTSys63104.2024.10780570.

[30] H. Kim, S. Lim, Temporal patternization of power signatures for appliance classification in nilm, Energies (Basel) 14 (2021), https://doi.org/10.3390/en14102931.

[31] Guizien V, Author M, Ubeda ES, Veiga A, Endesa S. Energy disaggregation and appliance detection in the context of NILM using Deep Learning. n.d.

[32] M. Kaselimi, N. Doulamis, Long-term recurrent convolutional networks for non-intrusive load monitoring. ACM International Conference Proceeding Series, ACM Trans. Math. Software (2020) 461–464, https://doi.org/10.1145/3389189.3397995.

[33] G.E. de Luis, Combating child sexual abuse 0n the internet: considerations in the light of the proposal for a regulation, Actualidad Juridica Iberoamericana (2024) 104–129. https://doi.org/10.13039/501100011033.

[34] H. Chen, Y.H. Wang, C.H. Fan, A convolutional autoencoder-based approach with batch normalization for energy disaggregation, J. Supercomput. 77 (2021) 2961–2978, https://doi.org/10.1007/s11227-020-03375-y.

[35] F. Zeng, J. Chu, C.T. SingleNet, An end-to-end single network for non-intrusive load monitoring of multiple appliances, in: Proceedings of 2020 IEEE International Conference on Artificial Intelligence and Computer Applications, ICAICA 2020, 2020, https://doi.org/10.1109/ICAICA50127.2020.9182604.

[36] Y. Yang, J. Zhong, W. Li, T. Aaron Gulliver, S. Li, Semisupervised multilabel deep learning based nonintrusive load monitoring in smart grids, IEEE Trans Industr Inform 16 (2020) 6892–6902, https://doi.org/10.1109/TII.2019.2955470.

[37] Z. Yan, D. Brunelli, D. Macii, M. Nardello, H. Wen, D. Petri, Appliance identification in NILM through image or signal features, a Performance Comparison. (2024). https://doi.org/10.23919/AEIT63317.2024.10736753.

[38] Wang Z, Zhang Z, Wang Y, Wen S, Liu Y, Ren B. Threshold Method Incorporated CNN Based Non-invasive Load Behavior Identification. 2024 IEEE IAS Industrial and Commercial Power System Asia, I and CPS Asia 2024, Institute of Electrical and Electronics Engineers Inc.; 2024, p. 561–6. https://doi.org/10.1109/ICPSAsia61913.2024.10761688.

[39] B. Zhao, X. Li, W. Luan, B. Liu, Apply graph signal processing on NILM: an unsupervised approach featuring power sequences, Sensors 23 (2023), https://doi.org/10.3390/s23083939.

[40] S. Sykiotis, M. Kaselimi, A. Doulamis, N. Doulamis, ELECTRIcity: an efficient transformer for non-intrusive load monitoring, Sensors 22 (2022), https://doi.org/10.3390/s22082926.

[41] M.H. Shahab, H.M. Buttar, A. Mehmood, W. Aman, M.M.U. Rahman, M.W. Nawaz, et al., Energy disaggregation & appliance identification in a smart home: transfer learning enables edge, Computing (2025), https://doi.org/10.1109/ISGTMiddleEast65737.2025.11314411.

[42] Senemmar S, Zhang J. Convolutional wavelet neural network based non-intrusive load monitoring for next generation shipboard power systems 2023. https://doi.org/10.36227/techrxiv.22820015.v3.

[43] F.N.S. Medeiros, G.L.B. Ramalho, M.P. Bento, L.C.L. Medeiros, On the evaluation of texture and color features for nondestructive corrosion detection, EURASIP J Adv Signal Process (2010; 2010.), https://doi.org/10.1155/2010/817473.

[44] D. Ding, J. Li, K. Zhang, H. Wang, K. Wang, T. Cao, Non-intrusive load monitoring method with inception structured CNN, Appl. Intell. 52 (2022) 6227–6244, https://doi.org/10.1007/s10489-021-02690-y.

[45] R. Zhang, Y. Sun, M.A. Mustafa, Proactive load-shaping strategies with privacy-cost trade-offs in residential households based on deep reinforcement learning, in: 2024 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids, SmartGridComm 2024, Institute of Electrical and Electronics Engineers Inc, 2024, pp. 160–166, https://doi.org/10.1109/SmartGridComm60555.2024.10738068.
[46] M. Yang, Z. Cheng, X. Liu, A non-intrusive load decomposition model based on multiple electrical parameters to point, Energies (Basel) 17 (2024), https://doi.org/10.3390/en17174482.
[47] Herath N, Herath M, Thilakanayake T, Angammana C, Liyanage M. An Image-Based Disaggregation Study of Time Series Energy Data Using Gramian Angular Field. Asia-Pacific Power and Energy Engineering Conference, APPEEC, IEEE Computer Society; 2023. https://doi.org/10.1109/APPEEC57400.2023.10561942.
[48] H. Hwang, S. Kang, Nonintrusive load monitoring using an LSTM with feedback structure, IEEE Trans. Instrum. Meas. 71 (2022), https://doi.org/10.1109/TIM.2022.3169536.
[49] K. Zhou, Z. Zhang, X. Lu, Non-intrusive load monitoring based on an efficient deep learning model with local feature extraction, IEEE Trans Industr Inform 20 (2024) 9497–9507, https://doi.org/10.1109/TII.2024.3383521.
[50] Cheng Y, Xu F, Li H, Chen Y, Xu C, Lin Y. Non-Intrusive Load Monitoring Based on Diffusion Model for Transfer Learning among Buildings. Proceedings - 2023 2nd Asian Conference on Frontiers of Power and Energy, ACFPE 2023, Institute of Electrical and Electronics Engineers Inc.; 2023, p. 156–60. https://doi.org/10.1109/ACFPE59335.2023.10455380.
[51] H. Yang, Z. Liang, X. Shi, J. Cheng, J. Liang, H. Dong, et al., Adversarial data anomaly detection and calibration for nonintrusive load monitoring, IEEE Internet Things J. 12 (2025) 544–555, https://doi.org/10.1109/JIOT.2024.3463730.
[52] Shahab MH, Mujtaba Buttar H, Mehmood A, Aman W, Mahboob M, Rahman U, et al. Transfer learning for non-intrusive load monitoring & appliance identification in a smart home. 2025. https://doi.org/10.1109/ISGTMiddleEast65737.2025.11314411.
[53] Li Y, Yin B, Wang P, Zhang R. Non-intrusive Load Monitoring Based on Convolutional Neural Network Mixed Residual Unit. J. Phys. Conf. Ser., vol. 1176, Institute of Physics Publishing; 2019. https://doi.org/10.1088/1742-6596/1176/2/022052.
[54] Putrada AG, Alamsyah N, Pane SF, Fauzan MN. GRU-MF: A Novel Appliance Classification Method for Non-Intrusive Load Monitoring Data. Proceeding - IEEE International Conference on Communication, Networks and Satellite, COMNETSAT 2022, Institute of Electrical and Electronics Engineers Inc.; 2022, p. 200–5. https://doi.org/10.1109/COMNETSAT56033.2022.9994409.
[55] X. Zhou, S. Li, C. Liu, H. Zhu, N. Dong, T. Xiao, Non-Intrusive load monitoring using a CNN-LSTM-RF model considering label correlation and class-imbalance, IEEE Access 9 (2021), https://doi.org/10.1109/ACCESS.2021.3087696.
[56] Herath M, Thilakanayake T, Angammana C, Liyanage M. Appliance Anomaly Detection as NILM Extention. Asia-Pacific Power and Energy Engineering Conference, APPEEC, IEEE Computer Society; 2023. https://doi.org/10.1109/APPEEC57400.2023.10561901.
[57] K. Li, B. Yin, Z. Du, Y. Sun, A nonintrusive load identification model based on time-frequency features fusion, IEEE Access 9 (2021) 1376–1387, https://doi.org/10.1109/ACCESS.2020.3047147.
[58] Z. Fang, D. Zhao, C. Chen, Y. Li, Y. Tian, Nonintrusive appliance identification with appliance-specific networks, IEEE Trans. Ind. Appl. 56 (2020) 3443–3452, https://doi.org/10.1109/TIA.2020.2994279.
[59] S.S. Hosseini, K. Agbossou, S. Kelouwani, A. Cardenas, N. Henao, A practical approach to residential appliances on-line anomaly detection: a case study of standard and smart refrigerators, IEEE Access 8 (2020), https://doi.org/10.1109/ACCESS.2020.2982398.
[60] Y.T. Quek, W.L. Woo, T. Logenthiran, A low cost master and slave distributed intelligent meter for non-intrusive load classification and anomaly warning, in: I2MTC 2018–2018 IEEE International Instrumentation and Measurement Technology Conference: Discovering New Horizons in Instrumentation and Measurement, 2018, https://doi.org/10.1109/I2MTC.2018.8409566.
[61] H. Rashid, P. Singh, V. Stankovic, L. Stankovic, Can non-intrusive load monitoring be used for identifying an appliance's anomalous behaviour? Appl. Energy 238 (2019) https://doi.org/10.1016/j.apenergy.2019.01.061.
[62] M. D'Incecco, S. Squartini, M. Zhong, Transfer learning for non-intrusive load monitoring, IEEE Trans. Smart Grid 11 (2020), https://doi.org/10.1109/TSG.2019.2938068.
[63] Y. Zhang, B. Yin, Y. Cong, Z. Du, Multi-state household appliance identification based on convolutional neural networks and clustering, Energies (Basel) 13 (2020), https://doi.org/10.3390/en13040792.
[64] D. de Paiva Penha, A.R. Garcez Castro, Home appliance identification for nilm systems based on deep neural networks, Int. J. Artif. Intell. Appli. 9 (2018) 69–80, https://doi.org/10.5121/ijaia.2018.9206.
[65] J.G. Kim, B. Lee, Appliance classification by power signal analysis based on multi-feature combination multi-layer LSTM, Energies (Basel) 12 (2019), https://doi.org/10.3390/en12142804.
[66] Zhang B, Luo F, He Y. Evaluation of Diffusion Models on Non-Intrusive Load Monitoring. 2023 IEEE 7th Conference on Energy Internet and Energy System Integration, EI2 2023, Institute of Electrical and Electronics Engineers Inc.; 2023, p. 4385–90. https://doi.org/10.1109/EI259745.2023.10512861.
[67] Shiddieqy HA, Hariadi FI, Adijarto W. Plug-load classification based on CNN from V-I trajectory Image using STM32. Proceeding - 2021 International Symposium on Electronics and Smart Devices: Intelligent Systems for Present and Future Challenges, ISESD 2021, Institute of Electrical and Electronics Engineers Inc.; 2021. https://doi.org/10.1109/ISESD53023.2021.9501919.
[68] I. Guzman, K. Goossen, K. Barner, Classification of high frequency NILM transients based on convolutional neural networks, in: 2022 IEEE Green Energy and Smart Systems, IGESSC 2022, Institute of Electrical and Electronics Engineers Inc, 2022, https://doi.org/10.1109/IGESSC55810.2022.9955332.
[69] S. Mari, G. Bucci, F. Ciancetta, E. Fiorucci, A. Fioravanti, An embedded deep learning NILM system: a year-long field study in real houses, IEEE Trans. Instrum. Meas. 72 (2023) 1–15, https://doi.org/10.1109/TIM.2023.3328085.
[70] D. Zhou, Y. Wu, H. Zhou, A nonintrusive load monitoring method for microgrid EMS using Bi-LSTM algorithm, Complexity (2021), https://doi.org/10.1155/2021/6688889.
[71] Z.M. Zhao, F. Wu, H. Wu, T. Liu, C. Zhou, J. Ma, et al., DCML: Boosting applying experience of NILM with dilated convolution and multi-task learning, in: 2024 IEEE/CIC International Conference on Communications in China, ICCC 2024, Institute of Electrical and Electronics Engineers Inc, 2024, pp. 991–996, https://doi.org/10.1109/ICCC62479.2024.10681804.
[72] X. Chang, H. Cui, M. Yang, Non-intrusive load decomposition algorithm based on temporal convolutional network and long short-term memory model combined with strongly correlated power sequences, Energy Rep. 9 (2023) 511–522, https://doi.org/10.1016/j.egyr.2023.04.163.
[73] M.H. Shahab, H.M. Buttar, A. Mehmood, W. Aman, M.M.U. Rahman, M.W. Nawaz, et al., Energy disaggregation & appliance identification in a smart home: transfer learning enables edge, Computing (2025), https://doi.org/10.1109/ISGTMiddleEast65737.2025.11314411.
[74] C.H. Hur, H.E. Lee, Y.J. Kim, S.G. Kang, Semi-supervised domain adaptation for multi-label classification on nonintrusive load monitoring, Sensors 22 (2022), https://doi.org/10.3390/s22155838.
[75] S. Ghosh, A. Mitra, S. Chakrabarti, A. Sharma, Decomposition-transformation assisted optimized heterogeneous classification strategy in NILM, IEEE Trans. Instrum. Meas. 72 (2023), https://doi.org/10.1109/TIM.2023.3291008.
[76] A. Faustine, L. Pereira, Multi-label learning for appliance recognition in NILM using fryze-current decomposition and convolutional neural network, Energies (Basel) 13 (2020), https://doi.org/10.3390/en13164154.
[77] A. Faustine, L. Pereira, Improved appliance classification in non-intrusive load monitoring usingweighted recurrence graph and convolutional neural networks, Energies (Basel) 13 (2020), https://doi.org/10.3390/en13133374.
[78] Faustine A, Pereira L. Applying Symmetrical Component Transform for Industrial Appliance Classification in Non-Intrusive Load Monitoring. ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings, vol. 2023- June, Institute of Electrical and Electronics Engineers Inc.; 2023. https://doi.org/10.1109/ICASSP49357.2023.10096324.
[79] A. Faustine, L. Pereira, H. Bousbiat, S. Kulkarni, UNet-NILM: A deep neural network for multi-tasks appliances state detection and power estimation in NILM, in: NILM 2020 - Proceedings of the 5th International Workshop on Non-Intrusive Load Monitoring, Association for Computing Machinery, 2020, pp. 84–88, https://doi.org/10.1145/3427771.3427859.
[80] Cavalca DL, Fernandes RAS. Recurrence plots and convolutional neural networks applied to nonintrusive load monitoring. IEEE Power and Energy Society General Meeting, vol. 2020- August, IEEE Computer Society; 2020. https://doi.org/10.1109/PESGM41954.2020.9281660.
[81] H. Yin, K. Zhou, S. Yang, Non-intrusive load monitoring by load trajectory and multi-feature based on DCNN, IEEE Trans Industr Inform 19 (2023) 10388–10400, https://doi.org/10.1109/TII.2023.3240924.
[82] L. De Diego-Oton, A. Hernandez, R. Nieto, M.C. Perez-Rubio, Comparison of neural networks for high-sampling rate NILM scenario, in: 2022 IEEE International Symposium on Medical Measurements and Applications, MeMeA 2022 - Conference Proceedings, Institute of Electrical and Electronics Engineers Inc, 2022, https://doi.org/10.1109/MeMeA54994.2022.9856406.
[83] F. Zhang, L. Qu, W. Dong, H. Zou, Q. Guo, Y. Kong, A novel NILM event detection algorithm based on different frequency scales, IEEE Trans. Instrum. Meas. 71 (2022), https://doi.org/10.1109/TIM.2022.3181897.
[84] Zhou Y, Ji C, Dong Z, Yang L, Zhou L. Non-intrusive Load Disaggregation Based on Multiple Optimization of Appliance Features and CNN-NLSTM Model. I and CPS Asia 2022 - 2022 IEEE IAS Industrial and Commercial Power System Asia, Institute of Electrical and Electronics Engineers Inc.; 2022, p. 871–6. https://doi.org/10.1109/ICPSAsia55496.2022.9949654.
[85] M. Zhou, S. Shao, X. Wang, Z. Zhu, F. Hu, Deep learning-based non-intrusive commercial load monitoring, Sensors 22 (2022), https://doi.org/10.3390/s22145250.
[86] Z. Zhou, Y. Xiang, H. Xu, Y. Wang, D. Shi, Unsupervised learning for non-intrusive load monitoring in smart grid based on spiking deep neural network, J. Mod Power Syst. Clean Energy 10 (2022) 606–616. https://doi.org/10.35833/MPCE.2020.000569.
[87] Liu X, Liao D, Zhang X, Gao H. Appliance Classification by Using Dynamic Voltage-Current Trajectory and 3D-Convolutional Neural Network. 2024 Boao New Power System International Forum - Power System and New Energy Technology Innovation Forum, NPSIF 2024, Institute of Electrical and Electronics Engineers Inc.; 2024, p. 468–74. https://doi.org/10.1109/NPSIF64134.2024.10883241.
[88] B. Liu, W. Liu, W. Luan, Y. Yu, B. Zhao, Y. Wang, Automatic appliance labeling for unsupervised NILM based on hierarchical decision-making, IEEE Trans. Instrum. Meas. 73 (2024) 1–13, https://doi.org/10.1109/TIM.2024.3381273.
[89] S. Sykiotis, S. Athanasoulias, M. Kaselimi, A. Doulamis, N. Doulamis, L. Stankovic, et al., Performance-aware NILM model optimization for edge deployment, IEEE Trans. Green Commun. Networking 7 (2023) 1434–1446, https://doi.org/10.1109/TGCN.2023.3244278.
[90] S. Athanasoulias, S. Sykiotis, M. Kaselimi, A. Doulamis, N. Doulamis, N. Ipiotis, OPT-NILM: an iterative prior-to-full-training pruning approach for cost-effective

user side energy disaggregation, IEEE Trans. Consum. Electron. 70 (2024) 4435–4446, https://doi.org/10.1109/TCE.2023.3324493.
[91] S. Athanasoulias, N. Temenos, N. Doulamis, A. Doulamis, I. Kokos, N. Ipiotis, Towards edge-computed NILM: insights from a Mediterranean use case, in: 2024 3rd International Conference on Energy Transition in the Mediterranean Area, SyNERGY MED 2024, Institute of Electrical and Electronics Engineers Inc, 2024, https://doi.org/10.1109/SyNERGYMED62435.2024.10799297.
[92] S. Nguyen, P. Shu-Ling Chen, Y. Du, Risk assessment of maritime container shipping blockchain-integrated systems: an analysis of multi-event scenarios, Transp Res E Logist Transp Rev 163 (2022), https://doi.org/10.1016/j.tre.2022.102764.
[93] Jia Y, Wang H, Batra N, Whitehouse K. A tree-structured neural network model for household energy breakdown. The Web Conference 2019 - Proceedings of the World Wide Web Conference, WWW 2019, Association for Computing Machinery, Inc; 2019, p. 2872–8. https://doi.org/10.1145/3308558.3313405.